

ECVOL-bench: An Open, Identity-Controlled Benchmark for Multimodal Earnings-Call Volatility Prediction

(anonymized working draft for comments — please do not cite or circulate)

Anonymous Author(s)

Draft compiled September 2, 2026

Abstract

A decade of work predicts post-earnings-call stock volatility from the text and audio of quarterly conference calls. We argue that the evaluation substrate this literature shares cannot support its claims: the canonical benchmarks are scored with a zero-day embargo against 30-day targets, with 44–80% of test calls from companies also present in training, on label files inherited unchanged from paper to paper, and under a horizon convention that downstream work misattributes *pending: regenerated by ecvol audit substrate*, Table 1]. We release ECVOL-bench, an open multimodal benchmark over FinCall-Surprise and MAEC that repairs this substrate: reconstructed company identity and call dates for a corpus that ships without them (92.9% of calls; 95.3% of earnings calls; 60/60 manual audit), three target families (level log-realised-volatility, change Δv , HAR-residual) under both the calendar-day and trading-day horizon conventions, temporal and ticker-disjoint splits with a 30-trading-day embargo, SHA-256 manifests, strong econometric baselines, and an identity-control suite (linear ticker probes, same-ticker and global content shuffles). Run over modern open-weight text and audio representations, the benchmark returns a consistent answer: text embeddings encode which company is speaking (89.5% ticker-probe accuracy; within-ticker transcript shuffles leave performance unchanged), frozen audio content is inert beyond past volatility (its apparent gains survive even a global audio shuffle), and fusion never significantly beats the best unimodal model—under honest controls, apparent content signal in this task family is overwhelmingly company identity plus past volatility. *pending: We then re-run the field’s five publicly runnable models on their own data under these controls (Table 8), re-evaluate the audio ladder on clean 2025 recordings, and evaluate the frozen pipeline on post-LLM-cutoff calls.* All code, configs, manifests, reconstructed metadata, split files, targets, derived features, and result tables are released.

1 Introduction

Earnings conference calls are one of the few recurring events at which management speaks at length, semi-

spontaneously, about a public company—in both *what* is said (the transcript) and *how* it is said (the audio). A line of research beginning with Qin and Yang (2019) and continuing through HTML (Yang et al., 2020), MAEC (Li et al., 2020), KeFVP (Niu et al., 2023), and ECC Analyzer (Cao et al., 2024a) claims that multimodal models over this data predict post-call stock volatility better than classical methods.

Two recent observations cast doubt on how much of that progress is real. First, Yu et al. (2024) (“Same Company, Same Signal”) show that transcript representations used across this literature predominantly encode *company identity* rather than content: same-ticker transcripts cluster tightly, and training-free baselines built on a company’s own past volatility beat all transcript-based models on their evaluation. Because volatility is strongly autocorrelated and company-specific, a model that learns “which company is this” inherits most of the apparent skill. Second, a 2025 survey of financial NLP (Tatarinov et al., 2025) finds that only $\sim 14\%$ of reported results reproduce exactly, attributing the gap to dead data links, API drift, and contamination.

This paper’s premise is that both problems are properties of the *evaluation substrate* the field shares, and that the substrate can be repaired. The lineage above evaluates on a small number of label and split files that are passed from paper to paper essentially unchanged. Those files (§3.1) impose no embargo between training and test windows even though targets run 30 days past each call; they place most test calls’ companies in the training set, and no paper reports the ticker-disjoint condition; they carry degenerate label cells; and they encode a horizon convention (calendar days) that the downstream literature describes as trading days. On such a substrate a model cannot be told apart from a memory of which company it is looking at. Meanwhile the corpus with the best license and the only raw audio, FinCall-Surprise (Shu et al., 2025), ships with *no* ticker, company, date, or speaker metadata at all—so the identity controls that would expose the confound cannot even be run on it without first reconstructing the identity the dataset omits.

We therefore build and release ECVOL-bench, an open benchmark that supplies what the substrate lacks, and we use it to re-examine the task’s central claims. The

benchmark is the contribution; the empirical results are the evidence for why it is needed. Concretely, we ask four research questions:

RQ1 Does earnings-call *content* (text and/or audio) add predictive signal for post-call volatility beyond past-volatility persistence (HAR-RV, GARCH)?

RQ2 Is multimodal (audio + text) better than the best unimodal model, after controlling for past volatility?

RQ3 Do the published models of this lineage, re-run on their own data, retain their reported advantage under embargoed, ticker-disjoint evaluation?

RQ4 Do any positive results survive (a) ticker-identity controls, (b) ticker-disjoint splits, and (c) evaluation on post-LLM-knowledge-cutoff calls?

The study was pre-registered so that no single outcome strands the work; the modelling stages, controls, and confirmatory tests were fixed before any model result existed and are recorded in a public decision log. RQ1, RQ2, and RQ4(a–b) are answered here; RQ3 and RQ4(c) are the subject of the reproduction and post-cutoff studies that this draft still owes (marked *[pending: like this]*).

Contributions.

1. **ECVOL-bench**, an open multimodal volatility benchmark over FinCall-Surprise (primary) and MAEC (secondary): reconstructed identity and dates for FinCall-Surprise (92.9% of calls; 95.3% of earnings calls; 60/60 manual audit), three target families (level v , change Δv , HAR-residual) under both the calendar-day and trading-day horizon conventions, temporal and ticker-disjoint splits with a 30-trading-day embargo, SHA-256 manifests, and a reproducible harness—released in full (§3).
2. **A substrate audit** of the label and split files the prior literature shares: embargo, ticker overlap, label defects, and the horizon convention, regenerable from one command (§3.1, Table 1).
3. **Honest baselines and an identity-control suite**—persistence, EWMA, HAR-RV, GARCH(1,1), and ticker-fixed-effect models for every horizon, split, and target; identity probes; transcript- and audio-shuffle controls; ticker-disjoint evaluation; a gender-confound analysis—and what they show when modern open-weight text and audio representations are run through them on consumer hardware (§6).
4. **A reproduction study** of the five publicly runnable models in this lineage, each re-run on its own original data under the new controls, together with a code-availability audit of the fifteen that cannot be run *[pending: Phase 6R; Table 8]*.

5. **Post-cutoff evaluation**: the audio ladder re-run on clean 2025 recordings, and the frozen pipeline evaluated on calls held after the knowledge cutoff of every model used *[pending: Phase 7 and T9.4]*.

2 Related Work

The benchmark lineage. Qin and Yang (2019) introduced text+audio volatility prediction and the 572-call EarningsCall dataset, establishing the 3/7/15/30-day log-realised-volatility target family that the field still uses (defined over *calendar* days in the original, a detail that matters in §3.1). MAEC (Li et al., 2020) scaled this to $\sim 3,443$ sentence-aligned multimodal calls over the S&P 1500. HTML (Yang et al., 2020) introduced a hierarchical transformer with multi-task learning across horizons; VolTAGE (Sawhney et al., 2020) added graph convolution over inter-stock relations with text/audio fusion; DialogueGAT (Sang and Bao, 2022) modelled speaker/dialogue structure with graph attention. KeFVP (Niu et al., 2023) combined knowledge-enhanced pre-training with a price-conditioned time-series module and is the standard SOTA reference. More recent work moves toward LLMs: ECC Analyzer (Cao et al., 2024a) uses hierarchical LLM summarisation with a RAG “question bank” and wav2vec2 features; RiskLabs (Cao et al., 2024b) fuses calls with news and market series; AT-FinGPT (Liu et al., 2025) is an audio-text LLM. Numeral-aware variants (NumHTML (Yang et al., 2022), NAM/ECNum (Chen et al., 2021a), GNAVol (Shi et al., 2023)) and Q&A-structure models (Ye et al., 2020) target specific aspects of call language. Of these, only a minority release code that still runs; §4 audits which.

The shared substrate. Nearly every paper above evaluates on label and split files derived from Qin and Yang (2019) and redistributed by successors (VolTAGE, KeFVP), rather than recomputing targets from prices. Yu et al. (2024) are the first to report that training-free baselines built on a company’s own past volatility beat every transcript model on these files, and release augmented earnings histories that make such baselines easy to build. No prior work audits the files themselves—embargo, train/test company overlap, label integrity, or the horizon convention—which is the gap §3.1 fills.

Audio and prosody. Chen et al. (2025) report that physics-informed acoustic features explain up to 43.8% of 30-day realised-volatility variance, motivating an audio ladder; the FinAudio benchmark (Cao et al., 2025) cautions that audio-LLMs struggle on long financial audio, informing our preference for frozen pooled features over end-to-end audio-LLMs in v1.

Known pitfalls (the methodological core). Our design is organised around four documented failure modes.

(1) *Ticker-identity leakage* (Yu et al., 2024) is the central threat to validity (§1); LLMs also deanonymise entity-neutered call transcripts with near-perfect accuracy (Lopez-Lira et al., 2025), so identity leaks even under masking. We respond with identity-robust targets (Δv , HAR-residual), a ticker-fixed-effect baseline, and an identity-control suite (§5). (2) *LLM lookahead bias*: a 2024-cutoff model “predicting” 2019–2021 outcomes it has effectively seen. A now-active literature offers diagnostics (Gao et al., 2025), cutoff-bounded pretraining (Yan et al., 2026; He et al., 2025), workflow benchmarks that find significant bias in open model families we use (Benhenda, 2026), and memorization filtering (Roy and Roy, 2026); the nearest precedent validates an LLM disagreement measure post-cutoff on ECB press conferences (Collodel, 2025). None evaluates a frozen earnings-call pipeline on post-cutoff calls; we respond with exactly that. (3) *Weak baselines*: much prior work omits persistence, GARCH(1,1), and HAR-RV (Corsi, 2009), which are hard to beat at long horizons; we compute these first and use persistence as the denominator of our headline metric (Gu et al., 2020). (4) *Gender bias in audio features* (Wang et al., 2024; Lin et al., 2024; Mathur et al., 2022): we analyse and report a gender confound rather than debias. We deliberately exclude agentic trading systems and time-series foundation models from v1 scope.

3 The ECVOL Benchmark

We first document the substrate the literature shares (§3.1), then describe what ECVOL-bench supplies in its place: corpora with reconstructed identity (§3.2), targets under both horizon conventions (§3.3), measured call timing (§3.4), leakage-proof splits (§3.5), and the release (§3.6).

3.1 The substrate the literature shares

The EarningsCall (EC) benchmark of Qin and Yang (2019) and the two MAEC releases (2015 and 2016 cohorts; Li et al., 2020) are evaluated, across the lineage in §2, on split and label files that are redistributed from paper to paper rather than recomputed. We fetched the canonical files from the VoltAGE, KeFVP, and HTML repositories, manifested them by SHA-256, and audited four properties. Table 1 (Result Table 5R) is regenerated from those copies by `ecvol audit substrate`; every figure below carries its source file and counts in the released CSV. (1) *Provenance*. The VoltAGE and KeFVP test-split files are byte-identical: the lineage shares one physical label set, never recomputed from prices. (2) *Embargo*. Every train/validation and validation/test boundary has a zero-day gap—on EC, training ends and validation starts on the same date (2017-08-03; 4 training and 14 validation calls on that day)—against targets that run 30 days past each call, so every boundary leaks the target window. (3) *Company overlap*. On

Benchmark / split	calls (tr/va/te)	overlap	embargo tr→
EC (Qin & Yang)	392 / 56 / 112	80.4%	
MAEC-15 (KeFVP files)	535 / 76 / 154	44.2%	
MAEC-16 (KeFVP files)	980 / 140 / 280	55.7%	
ours: FinCall temporal	1480 / 228 / 481	91.3%	48
ours: FinCall ticker-disjoint	1701 / 239 / 484	0.0%	
ours: MAEC temporal	1604 / 73 / 507	78.9%	101
ours: MAEC ticker-disjoint	1805 / 259 / 514	0.0%	r
ours: Earnings25 ticker-disjoint	333 / 47 / 95	0.0%	r

Table 1: Substrate audit of the shared benchmark files (Result Table 5R, regenerated by `ecvol audit substrate` from SHA-256-manifested copies of the VoltAGE and KeFVP repositories). The VoltAGE and KeFVP split files are byte-identical. Overlap is a property every temporal split permits, ours included; the point is that no paper in the lineage reports the disjoint condition. The zero-day embargos leak 30-day target windows across every boundary. Shipped MAEC labels join our calendar-day targets on 74–77% of test calls at $\tau \geq 7$ and correlate 0.51–0.88 with them (0.25–0.71 at $\tau=3$).

the temporal splits the field reports, 80.4% (EC), 44.2% (MAEC-15), and 55.7% (MAEC-16) of test calls come from companies that also appear in training. A temporal split *always* permits this; the finding is not that overlap exists but that no paper in the lineage reports the disjoint condition, so the identity confound of Yu et al. (2024) has never been measured on these benchmarks. (4) *Label integrity and convention*. The single-day log-vol series contain 137 exact-zero cells out of 16,800 (0.82%; the headline averaged series is clean), and KeFVP’s MAEC files interleave four binary labels at exactly $\tau \in \{3, 7, 15, 30\}$ among 26 price columns under identical `future_label_N` naming. Finally, the source paper defines its horizons over *calendar* days; most downstream work, and our own design document until its correction, describes them as trading days—at $\tau=30$ the difference is roughly 21 versus 30 sessions, and every published number depends on which one was used. (5) *Labels versus prices*. KeFVP’s shipped MAEC test labels key on our own call identifiers; on the 74–77% of calls we can price at $\tau \geq 7$ they correlate only 0.51–0.88 with our calendar-day targets (0.25–0.71 at $\tau=3$, where half the windows hold one session), with mean absolute log-vol differences of 0.15–0.73—so the lineage’s labels are not reproducible from prices under any single documented rule.

3.2 Corpora and identity reconstruction

We build on two openly licensed multimodal earnings-call corpora and reserve a third, post-cutoff corpus for the lookahead and clean-audio experiments. **FinCall-Surprise** (Shu et al., 2025) (primary; Apache-2.0) contains 2,688 calls spanning 2019–2021 with full MP3 audio, JSON transcripts, and slides. **MAEC** (Li et al., 2020) (secondary; CC-BY-SA-4.0) contains ~3,443 calls over 2015–2018 (S&P 1500) with sentence-

aligned audio features and transcripts but no raw audio. **Earnings25** [TODO: cite; Zenodo 10.5281/zenodo.18762168] (CC-BY-4.0; ~500 Q4-2025 S&P 500 calls, 498 h of audio, held January–February 2026) is post-cutoff for every model we use and carries native audio at source quality; it is the Phase-7 corpus (§5) [pending: ingestion gated on a metadata check].

FinCall-Surprise ships with four fields per record and *no* ticker, company, CIK, date, time, fiscal period, or speaker names. We reconstruct identity and date from transcript content and slide metadata, cross-check against price data, and release the resulting table together with an 80-row manual override list: the reconstruction resolves 2,496/2,688 calls (92.9% overall; 95.3% of the 2,499 earnings-type calls) and passes a 60/60 manual audit. This table is what makes the identity controls of §5 runnable on the corpus at all.

3.3 Price data and targets

Daily adjusted-close OHLCV is obtained from a free, no-account source and cross-checked against Tiingo on a random ticker sample (adjusted-return correlation > 0.999); delisted or unmatched tickers produce an explicit coverage report rather than being silently dropped. After price joining, FinCall-Surprise yields 2,424 calls with at least one target (97.1% join rate) and MAEC yields 2,578 (75.4%; a documented shortfall driven by 2015–2018 delistings), for a combined modelling set of 5,002 calls.

For each call we define returns $r_t = (P_t - P_{t-1})/P_{t-1}$ relative to day $t = 0$, the last trading day before the call’s information is public (§3.4). The primary target is log realised volatility over horizon $\tau \in \{3, 7, 15, 30\}$,

$$v_{\text{post}}(\tau) = \ln\left(\sqrt{\frac{1}{\tau} \sum_{t=1}^{\tau} (r_t - \bar{r})^2}\right), \quad (1)$$

with pre-call volatility $v_{\text{pre}}(\tau)$ defined over the sessions ending at day 0. We ship every target under *both* horizon conventions. Under the **trading-day** convention the window is sessions $1..\tau$ ($n = \tau$). Under the **calendar-day** convention of Qin and Yang (2019), which the published leaderboard inherits, the window is every session dated within τ calendar days of day 0, so $n \approx 0.7\tau$ and varies with the weekday and holidays (FinCall: $n = 2.4/4.9/10.7/21.0$ sessions on average at $\tau = 3/7/15/30$). In both cases the variance is taken over the n sessions actually present, and a window holding fewer than two sessions is excluded with a reason code rather than reported as a one-return “volatility”—which removes **48.8%** of FinCall’s joined calls at $\tau=3$ calendar days (every Thursday or Friday call). Each convention is a separate file with an identical schema, a `convention` column and the per-row session counts; the trading-day file is the validated set every result in §6 uses, and every result table carries the convention as a column (§6.6). We add two identity-robust variants

the **change** $\Delta v(\tau) = v_{\text{post}}(\tau) - v_{\text{pre}}(\tau)$, which subtracts the company’s own level, and the **HAR-residual** $v_{\text{post}}(\tau) - \overline{\text{HAR}}(\tau)$, deviation from a train-only HAR-RV (Corsi, 2009) forecast. Insufficient post-call history and zero-variance windows yield NaN targets excluded with a reason code; targets are unit-tested against hand-computed values.

3.4 Call timing

Fewer than 4% of FinCall-Surprise transcripts mention a clock time, and 32.4% of recovered *dates* come from slide-PDF creation stamps, so no call in either corpus carries a measured time of day. All results in §6 therefore use a uniform, flagged after-hours rule (calls assumed to occur after market close; $t=1$ is the next trading session). We retrofit the information boundary per call from EDGAR: the acceptance time of the earliest 8-K carrying Item 2.02 (the earnings release) within three days before to one day after the call date, taken from the SEC submissions API and cached; where no such filing exists we fall back, in order, to Earnings25’s call-start time, to the before/after-market attribute of Yu et al. (2024), and finally to the after-hours assumption, with the tier recorded per call. Coverage by the EDGAR tier is 89.3% for FinCall-Surprise, 68.6% for MAEC, and 99.8% for Earnings25. Under the measured anchor, **53% of FinCall and 59% of MAEC calls change day 0**, almost all one session earlier: releases before the open are the norm (FinCall 53%, Earnings25 72%), so the uniform after-hours rule places the reaction session in the *pre* window for most of the corpus. Because every released label set in this literature anchors on the call date, this affects the published targets as much as ours. We keep the after-hours targets as the primary set for every result in §6 and ship the measured-anchor targets beside them (`targets_measured`), with a sensitivity table (`timing_sensitivity.csv`): persistence MSE rises 1–8% on FinCall and up to 26% on MAEC at $\tau=3$ once the reaction session enters the post window, while the HAR-RV R_{OOS}^2 moves by at most ± 0.08 ; the target correlation between anchors is 0.97 for v_{post} and 0.80 for Δv on Earnings25. Whether to re-baseline on the measured anchor is a release decision we flag rather than take here, since it moves every split boundary.

3.5 Leakage-proof splits

Three split families are generated once as committed CSVs and guarded by leakage assertions in CI. *Temporal*: train < validation < test in strict calendar order (~70/10/20) with a **30-trading-day embargo** between segments so that no 30-day target window crosses a boundary—the constraint the shared substrate lacks (§3.1). *Ticker-disjoint*: companies in validation/test never appear in training (grouped, seeded), the headline identity-control split; by construction its test-call com-

pany overlap is 0.0%, against 91.3% (FinCall-Surprise) and 78.9% (MAEC) on our own temporal splits. *Combined*: both constraints simultaneously, reported as the hardest robustness row. An information rule is codified and enforced: every feature function receives an `as_of` timestamp, and the harness fails any feature that reads price data after it. FinCall-Surprise spans COVID; per-year breakdowns are mandatory and a Feb–Apr 2020 exclusion is reported as an exploratory sensitivity.

3.6 Release

ECVOL-bench is released in full: the identity/date reconstruction table and override list, split CSVs, targets under both conventions, the derived-feature archives of §4, the evaluation harness and baselines, the substrate-audit report, and every result table. FinCall-Surprise-derived artifacts are Apache-2.0; MAEC-derived artifacts inherit CC-BY-SA-4.0. Raw audio and transcripts from sources with unclear licenses are never redistributed (*scripts-not-data* policy).

4 Models

We organise our own models as a ladder (Stages 0–4, Figure 1). Two invariants make the comparison honest. First, from Stage 2 upward every learned model includes past-volatility covariates (v_{pre} at all horizons plus the HAR-RV forecast) and is *also* reported with them ablated, baking the persistence control into every comparison. Second, given the small-data regime (10^3 – 10^4 samples) we use frozen representations with light heads (ridge, shallow MLP, LightGBM) and per-horizon models by default; fine-tuning is gated to a conditional later stage and a shared multi-task head is an ablation, not the headline. Because the shallow-MLP heads proved seed-unstable (across-seed R_{OOS}^2 std up to ≈ 3.4 ; see Limitations), the consolidated grid (Table 6) uses one pre-registered deterministic ridge/structural canonical model per stage; MLP heads appear in the appendix only.

Stage 0 — econometric baselines. Persistence ($\hat{v} = v_{\text{pre}}$), EWMA (RiskMetrics $\lambda=0.94$), HAR-RV (OLS on daily/weekly/monthly realised-variance factors, Corsi, 2009), and GARCH(1,1) (per-call fit, `arch`). These establish the floor and the denominator of the headline metric.

Stage 1 — tabular identity baseline. LightGBM over past-volatility features, trivial metadata (transcript length, Q&A turn count where available), and a **ticker fixed effect**. This is the in-house “Same-Company-Same-Signal” control: it measures how much apparent multimodal performance is identity plus metadata.

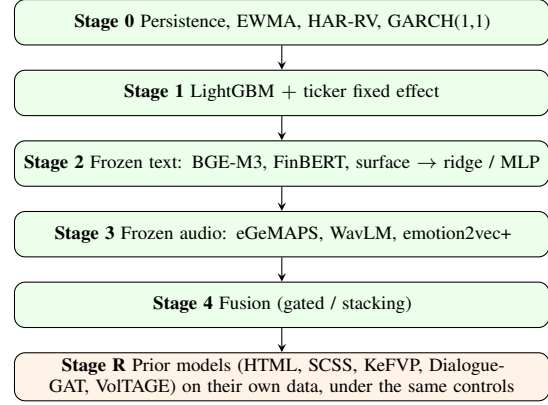


Figure 1: The modelling ladder. From Stage 2 up, every model also carries past-volatility covariates (and is reported with them ablated). Every stage runs through the identity-control suite. Green = complete, orange = pending.

Stage 2 — frozen text content. We segment each transcript into prepared-remarks vs. Q&A and speaker turns, then extract three feature blocks: BGE-M3 (BAAI, 2024) section-pooled sentence embeddings (1024-d), FinBERT (ProsusAI, 2019) sentiment aggregates per scope×role, and surface statistics. Heads are ridge (with the α grid selected on validation, *no* retrain on train+val) and a shallow MLP (1×256 , L2 + early stopping, 5 seeds). Features use fp32 with content-hash caching for bit-identical re-extraction.

Stage 3 — frozen audio content. Audio is QC’d and resampled to 16 kHz mono FLAC. We extract eGeMAPSv02 functionals (88-d, openSMILE (Eyben et al., 2010), CPU, interpretable), then WavLM-Large (Chen et al., 2021b) and emotion2vec+ (Ma et al., 2023) embeddings over 30 s non-overlapping windows, mean-pooled to a per-call vector (2,671 calls, all extracted on one consumer GPU with resumable content-hash caches), fed to the same heads. We use per-call pooling only (no per-sentence alignment), and report a gender-confound analysis.

Stage 4 — fusion. Gated fusion (per-modality train-fit PCA + norm gates → shallow MLP) and late-fusion stacking (a meta-ridge over the text, audio, and Stage-1 GBDT base predictions, fit on validation only) over frozen modality embeddings, to test whether multimodal beats the best unimodal model (RQ2). Cross-attention is omitted: with one pooled vector per modality, attention over 2–3 tokens reduces to gated fusion. All fusion heads are <5M parameters.

Reproduced prior models. Of ~ 20 papers in the lineage of §2, five release code that still runs: HTML (Yang et al., 2020), Same-Company-Same-Signal (Yu et al., 2024), KeFVP (Niu et al., 2023), DialogueGAT (Sang and Bao, 2022), and VoltAGE (Sawhney et al., 2020). We re-run each *on its own original dataset*,

first reproducing the published-split number within a stated tolerance, then evaluating the same model under an embargoed and a ticker-disjoint split of that dataset, so that “the model fails” cannot be confused with “the model does not transfer” (RQ3). The remaining fifteen—among them the strongest recent positive claims (Cao et al., 2024a; Chen et al., 2025)—have no code, empty or unpushed repositories, closed-API dependencies, or a stated inability to run; we tabulate each with the evidence [pending: Phase 6R; Table 8 and the code-availability audit table in the appendix].

5 Experimental Protocol

Metrics. We report MSE on log-RV per horizon (literature comparability); R_{OOS}^2 vs. persistence (Gu et al., 2020; $1 - \sum(y - \hat{y})^2 / \sum(y - \hat{y}_{\text{persist}})^2$), the headline metric, which is not gameable by identity memorisation the way raw MSE is; MAE; and cross-sectional Spearman rank correlation per calendar quarter. All metrics are reported on all three targets (v , Δv , HAR-residual) $\times$ all splits, and carry the horizon convention as a column.

Statistical rigor. Every learned model is run with ≥ 5 seeds (mean $\pm$ std). We use Diebold–Mariano tests (Diebold and Mariano, 1995) (HLN small-sample correction) on per-call squared-error differentials vs. persistence and vs. the best preceding stage; cluster-bootstrap confidence intervals (1,000 resamples) clustered by ticker and by calendar quarter; and Holm-corrected p -values across horizons for confirmatory claims. Confirmatory vs. exploratory analyses are labelled in advance: confirmatory analyses are Stage- k vs. Stage-1 DM tests on Δv at each τ , Stage 4 vs. best unimodal, the temporal-vs-ticker-disjoint gap per stage, and—for each reproduced prior model—the published-split vs. leakage-proof-split gap; all per-year/regime, COVID-exclusion, prepared-vs-Q&A, call-length, and gender-confound analyses are exploratory.

Identity-control suite. The signature diagnostic of this benchmark probes whether content models read content or identity, via four controls. (1) *Ticker-only model*: predict the per-ticker target mean (global mean for unseen tickers); if a content model matches it on level- v , that model reads identity. (2) *Same-ticker transcript shuffle*: replace each test call’s transcript with a different call from the same ticker; if performance is unchanged the model reads identity, not content, whereas a global shuffle (any other call) isolates pure overfitting. (3) *Identity probe*: a linear probe predicting the ticker from frozen call embeddings; high accuracy demonstrates identity is encoded. (4) *Ticker-disjoint split results* reported beside temporal-split results in every table. The suite runs unchanged on the reproduced prior models (§4).

Post-cutoff experiments. Earnings25 (§3.2) supports two experiments that the 2019–2021 corpora cannot. *Clean audio*: the frozen Stage-3 audio ladder (same extractors, same heads, same controls) is re-run on source-quality 2025 recordings, to test whether the audio-inert finding of §6 is a property of the task or of FinCall-Surprise’s uniform 40 kbps transcodes. *Lookahead*: the entire frozen pipeline—no retraining, no post-hoc threshold changes (rule fixed in advance)—is evaluated on calls held after the knowledge cutoff of every model used; disproportionate degradation relative to the 2019–2021 results would be contamination evidence. [pending: Both are owed: T9.4 and Phase 7.]

Reproducibility. All result tables regenerate byte-identically from committed CSV artifacts via `ecvol report`, guarded by a CI test. Every result CSV carries a committed run manifest—the resolved command config and its hash, git SHA, seeds, environment fingerprint, and the file’s SHA-256—and both `ecvol report` and CI refuse a result file whose bytes match no manifest; one committed config per result-producing command is the record of how it was run. Every data file carries a SHA-256 manifest entry; targets are unit-tested against hand-computed values; and the splits are guarded by leakage assertions in CI. Manifests written for results that predate this mechanism are labelled *backfill* until their producing command is re-run. [pending: Clean-machine regeneration of Table 2 is recorded with the reproducibility package.]

6 Results

We report results factually and defer interpretation to §7. Curated headline tables appear here on the identity-robust Δv target for FinCall-Surprise under the trading-day convention; the complete grid (both corpora, level- v and Δv , R_{OOS}^2 and MSE, all splits and horizons) is in Appendix A. Stars denote Diebold–Mariano significance ($p < 0.05$): in Table 2 vs. persistence, and in Tables 3 and 5 vs. the Stage-1 ticker-FE GBDT; in the consolidated grid (Table 6) markers are Holm-corrected across horizons and signed (* better / † worse than Stage-1).

6.1 Baselines (Stages 0–1)

Table 2 shows the econometric floor. HAR-RV is strong at short and medium horizons on both splits, confirming RQ1’s premise that past-volatility baselines are hard to beat. The one striking failure is the FinCall temporal split at $\tau=30$, where every model falls below persistence (HAR-RV $R_{\text{OOS}}^2 = -0.287$): the training window is COVID-inflated (Feb–May 2020) while the test window is calm late 2021, so a long-horizon level forecast trained on the crisis regime is mis-calibrated. The same models pass comfortably on the ticker-disjoint split (HAR-RV $R_{\text{OOS}}^2 = +0.229$ at $\tau=30$) and on

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
<i>Temporal split (n=481)</i>				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.387*	0.362*	0.145*	-0.219*
HAR-RV	0.464*	0.410*	0.207*	-0.287*
GARCH(1,1)	0.265*	0.193*	-0.140*	-0.541*
GBDT (ticker FE)	0.344*	0.248*	-0.057	-0.904*
<i>Ticker-disjoint split (n=484)</i>				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.357*	0.199*	-0.009	0.046
HAR-RV	0.495*	0.357*	0.211*	0.229*
GARCH(1,1)	0.356*	0.229*	0.088	0.209*
GBDT (ticker FE)	0.407*	0.275*	0.089	0.108

Table 2: Baselines. R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). HAR-RV is strong except on the COVID-regime temporal $\tau=30$ cell. *: Diebold–Mariano significant vs. persistence ($p < 0.05$).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
<i>Temporal split (n=481)</i>				
Ridge (text)	0.028*	0.030*	0.015	-0.144*
Ridge (past-vol)	0.415*	-1.053	-2.858	-0.667
Ridge (text+vol)	-1.678	-3.076	-3.586	-0.468
MLP (text)	-0.108*	-0.175*	-0.265*	-1.042
<i>Ticker-disjoint split (n=484)</i>				
Ridge (text)	-0.044*	-0.026*	0.008	0.174
Ridge (past-vol)	0.451	0.306	0.191*	0.224
Ridge (text+vol)	0.223*	0.129*	0.094	0.236
MLP (past-vol)	0.492*	0.363*	0.230*	0.249*

Table 3: Text content (Stage 2). R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). The text-only head does not separate from the past-volatility head; adding text to past-vol does not help. *: Diebold–Mariano significant vs. the Stage-1 ticker-FE GBDT ($p < 0.05$). Full head set in Appendix A.

6.2 Text content (Stage 2)

Table 3 shows frozen-text heads against the Stage-1 baseline on Δv . The result is a clean negative: *text content does not cleanly beat the identity/past-volatility baselines*. On the FinCall temporal split the text ridge head is near zero and collapses at $\tau=30$; the MLP heads overfit badly. On the ticker-disjoint split the text ridge head is weak (R^2_{OOS} between -0.04 and $+0.17$) while the past-vol-only head is consistently strong ($R^2_{\text{OOS}} \approx 0.19$ – 0.45), so any apparent text gain is not separable from what past volatility already provides. MAEC shows the same pattern at smaller magnitude (Appendix A).

Corpus	Identity probe (linear)	
	Accuracy	$\times$ chance
FinCall ($n=2656$, 357 tickers)	89.5%	319 $\times$
MAEC ($n=3132$, 902 tickers)	53.9%	487 $\times$

Condition	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Real transcript	0.028	0.030	0.015	-0.144
Within-ticker shuffle	0.017	0.020	0.006	-0.142
Global shuffle	-0.099	-0.121	-0.115	-0.266

Table 4: Identity controls. Top: a linear probe predicts the ticker from frozen BGE-M3 call embeddings far above chance, demonstrating identity is heavily encoded. Bottom: R^2_{OOS} of the text ridge head on the Δv target (FinCall, temporal split, test, $n=481$). Replacing each test call’s transcript with a *different call from the same ticker* (within-ticker shuffle) barely changes performance, whereas a global shuffle degrades it—evidence the head reads ticker identity rather than call content.

6.3 Identity controls (Stage 2 diagnostic)

Table 4 explains the negative result. A linear probe recovers the ticker from frozen call embeddings at 89.5% on FinCall (319 $\times$ chance) and 53.9% on MAEC (487 $\times$ chance): identity is heavily encoded. The transcript shuffle is decisive—replacing each test call’s transcript with a *different call from the same ticker* barely changes R^2_{OOS} , whereas a global shuffle degrades it. The heads are reading *which company* is speaking, not *what* is said. This is the in-corpus confirmation of Yu et al. (2024) and the reason RQ4 is central—and it is only measurable because the benchmark reconstructed the identity the corpus omits (§3.2).

6.4 Audio content (Stage 3)

Table 5 extends the same design to audio. Audio-only heads (eGeMAPS, WavLM-Large, emotion2vec+, ridge or MLP) sit at R^2_{OOS} zero on every split and horizon (-0.06 to $+0.03$): frozen audio content alone carries no measurable Δv signal. Adding past-volatility covariates reproduces the Stage-2 pattern—the only DM-significant wins over Stage-1 are past-vol-augmented heads on the ticker-disjoint split. The audio-shuffle control (bottom block) is decisive: replacing each test call’s audio with a random *other company’s* call leaves most of those gains intact ($\tau=3$: $0.337 \rightarrow 0.318$; $\tau=30$: $0.212 \rightarrow 0.100$), so the audio features are inert and past volatility carries the signal. Identity is nonetheless encoded: a linear probe recovers the ticker from per-call WavLM embeddings at 76.3% (271 $\times$ chance; emotion2vec+ 29.4%, 105 $\times$). An exploratory gender-confound check (eGeMAPS-F0 dominant-speaker proxy, 100% coverage) finds near-identical per-group test MSE (0.241 low-pitch vs. 0.242 high-pitch) and a 0.03 correlation between speaker F0 and squared error: with inert audio features, no large gender disparity is detectable (see the Ethics Statement

Model	$\tau=3$	$\tau=7$	$\tau=15$	Stage (canonical model)	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
<i>Temporal split (n=481)</i>				<i>Temporal split (n=481)</i>				
Ridge (eGeMAPS)	-0.000*	-0.007*	-0.015	S1 HAR-RV	0.464	0.410	0.207	-0.287
Ridge (WavLM)	-0.005*	-0.007*	-0.045	S1 GBDT (ticker FE)	0.344	0.248	-0.057	-0.904
Ridge (emotion2vec+)	-0.018*	-0.028*	-0.056	S1 Text (ridge+vol)	-1.678	-3.076	-3.586	-0.468
Ridge (WavLM+past-vol)	0.265	-1.026	-2.032	S1 Audio (ridge+vol)	0.214	-1.099	-3.706	-0.548
<i>Ticker-disjoint split (n=484)</i>				S4 Fusion (stack)	0.232	-0.326	-1.828	-0.010*
<i>Ticker-disjoint split (n=484)</i>				<i>Ticker-disjoint split (n=484)</i>				
Ridge (eGeMAPS)	-0.011*	-0.013*	-0.002	S2 HAR-RV	0.495	0.357	0.211	0.229
Ridge (WavLM)	-0.038*	-0.035*	-0.027*	S2 GBDT (ticker FE)	0.407	0.275	0.089	0.108
Ridge (emotion2vec+)	-0.002*	-0.003*	0.011	S2 Text (ridge+vol)	0.223†	0.129†	0.094	0.236
Ridge (WavLM+past-vol)	0.337*	0.171*	0.111	S2 Audio (ridge+vol)	0.304†	0.147†	0.044	0.209
<i>Audio-shuffle control of WavLM+past-vol (ticker-disjoint)</i>				S4 Fusion (stack)	0.436	0.292	0.133	0.244*
Real audio	0.337	0.171	0.111					
Within-ticker shuffle	0.317	0.144	0.097					
Global shuffle	0.318	0.115	0.093					

Table 5: Audio content (Stage 3). R_{OOS}^2 vs. persistence on the Δv target, FinCall-Surprise (test). Audio-only heads sit at zero everywhere; the apparent gains of the past-vol-augmented head survive replacing every test call’s audio with a *random other company’s* call (global shuffle, bottom block), so past volatility, not audio content, carries them. *: Diebold–Mariano significant vs. the Stage-1 ticker-FE GBDT ($p < 0.05$; read direction from the value). Full head set in Appendix A.

Table 6: Consolidated grid (Stages 0–4), pre-registered canonical model per stage. R_{OOS}^2 vs. persistence on the Δv target, FinCall-Surprise (test); persistence is 0 by construction. Significance is the Holm-corrected confirmatory family (Stage- k vs. Stage-1 DM across horizons): * = significantly better than Stage-1, † = significantly worse ($p < 0.05$). No content stage beats the econometric floor on the temporal split (the fusion $\tau=30$ star beats only the collapsed GBDT, not persistence); the ticker-disjoint gains are past-vol/identity-confounded (Table 5, bottom block). Canonical models are deterministic ridge/structural heads; seed-unstable MLP heads are reported in Appendix A only.

for caveats). All of this rests on FinCall-Surprise’s 40 kbps source audio; §6.8 is the test of whether it transfers.

6.6 Horizon convention: calendar vs. trading days

6.5 Fusion and the consolidated grid (Stage 4)

Gated fusion and late-fusion stacking over all frozen modalities answer RQ2 in the negative: *multimodal is not better than the best unimodal model*. Only one of 48 Δv test cells beats Stages 1–3 DM-significantly (gated fusion, temporal $\tau=7$, $R_{\text{OOS}}^2 = +0.019$ —negligible) and the best fusion cell overall (stacking, ticker-disjoint, $\tau=3$, $R_{\text{OOS}}^2=0.436$) is not significant vs. Stage-1 ($p=0.13$). Table 6 consolidates the ladder with one pre-registered canonical model per stage and Holm-corrected confirmatory tests. Two facts summarise it. On the regime-honest *temporal* split, no content stage beats the econometric floor at any horizon; where a content model is Holm-significantly better than Stage-1 (fusion at $\tau=30$) it merely beats a collapsed GBDT while remaining below persistence. On the *ticker-disjoint* split the favorable cells are exactly the past-vol-augmented ones that the shuffle controls (Tables 4 and 5) show to be identity/past-vol-confounded. The per-year breakdown sharpens the regime point: the FinCall temporal test year is entirely 2021, and there every content model underperforms persistence on level- v (HAR -0.29 , GBDT -0.82 , text -1.09 , audio -1.66 , fusion -1.02).

Table 7 reports the delta between the trading-day targets used above and the calendar-day targets of Qin and Yang (2019). Three facts matter for anyone comparing to the published leaderboard. First, the calendar window holds about 0.7τ sessions, and at $\tau=3$ it holds at most one session for any Thursday or Friday call: 48.8% of FinCall’s calls have no defensible $\tau=3$ calendar-day target and are excluded under our rules, whereas the released label sets carry a value for every call. Second, the two conventions are far from interchangeable at short horizons (correlation of v_{post} 0.75 at $\tau=3$, 0.93–0.96 beyond), and the persistence MSE—the leaderboard’s metric—roughly doubles at $\tau=3$ under calendar days because the target is averaged over two returns instead of three. A published MSE therefore carries no information about model quality unless the convention is stated. Third, the one cell where our baselines fail the pre-registered sanity gate (HAR-RV below persistence at $\tau=30$ on FinCall’s temporal split, the COVID-regime cell) becomes a marginal pass under the calendar convention ($R_{\text{OOS}}^2 -0.287 \rightarrow +0.017$), because a 21-session window straddles less of the regime shift than a 30-session one. The Stage-0/1 grid is shipped under both conventions (result_table_1, convention column); the Stage-2–4 content models are evaluated under the trading-day convention only, since the delta above is a property of the target, not of the head.

	$\tau=3$	Condition (EC)	$\tau=15$	$\tau=30$	model	$\tau=30$
<i>Targets (FinCall, calls ok under both conventions)</i>						
Sessions in calendar window (mean)	2.4	paper, published split, published labels	4.9	10.7	HTML (text, WWM-BERT)	1.175
Calls with ≥ 2 sessions (of 2,424)	1,241	published split, published labels	2,424	2,424	HTML head (BGE-M3)	0.769
$\text{corr}(v_{\text{post}}^{\text{trad}}, v_{\text{post}}^{\text{cal}})$	0.748	published split, our labels	0.925	0.957	persistence	1.485
$\text{corr}(\Delta v^{\text{trad}}, \Delta v^{\text{cal}})$	0.761	published split, our labels	0.895	0.909	HTML head (BGE-M3)	0.739
<i>Persistence MSE, level-v, temporal test (trading / calendar)</i>						
FinCall	1.174 / 2.307	embargoed temporal, our labels	0.419 / 0.597	0.208 / 0.288	HTML head (BGE-M3)	0.745
MAEC	1.383 / 2.391	ticker-disjoint, our labels	0.571 / 0.780	0.318 / 0.407	persistence	1.946
<i>HAR-RV R_{OOS}^2 vs. persistence, level-v, temporal test (trading / calendar)</i>						
FinCall	0.464 / 0.380	embargoed temporal, our labels	0.410 / 0.448	0.207 / 0.323	HTML head (BGE-M3)	0.808
MAEC	0.507 / 0.574	ticker-disjoint, our labels	0.392 / 0.454	-0.287 / 0.017	persistence	1.670

Table 7: Horizon convention delta. The calendar-day windows of [Qin and Yang \(2019\)](#) hold $\sim 0.7\tau$ sessions; at $\tau=3$ half the calls have at most one post-call session and are excluded. Persistence MSE (the leaderboard’s metric) roughly doubles at $\tau=3$ under the calendar convention, so published MSE values are not comparable across conventions; the FinCall temporal $\tau=30$ sign flip of HAR-RV (the COVID-regime cell, §6) is itself convention-dependent. Full model $\times$ horizon views under both conventions: data/results/result_table_1.tex.

6.7 Prior models under the new controls (Stage R)

Table 8 reports the first entry of the reproduction study: the HTML sentence-level transformer with its multi-task loss and published hyper-parameters ([Yang et al. 2020](#)), run over our frozen BGE-M3 chunk sequences (a documented encoder substitution; the head and protocol are the paper’s) on EC. Three things follow. First, the published number is only partly reproduced: at $\tau=3$ the head beats it (0.77 vs. 1.18), but the published $\tau=15$ and $\tau=30$ values (0.153, 0.133) are *below the persistence floor of the shipped labels* (0.340, 0.202) and no configuration we ran approaches them—the head is worse than persistence at $\tau=30$ in every condition, as every model in §6 is on FinCall. Second, label source barely matters (0.77/0.44/0.29/0.22 vs. 0.74/0.40/0.28/0.22), so the convention and anchor issues of §3.1 do not explain the shortfall. Third, the ticker-disjoint split raises the head’s MSE by 9% / 23% / 46% / 36% over the published split with identical labels, collapsing its R_{OOS}^2 against persistence at $\tau=15$ from 0.21 to 0.04, whereas the 30-session-embargoed temporal split costs nothing—the company-overlap confound, not the embargo, is what the published protocol leaves unmeasured. The second entry is Same-Company-Same-Signal ([Yu et al. 2024](#)) on its own DEC data: its training-free PEV and STPEV baselines reproduce to three decimals from the shipped file (2023 quarterly averages 0.240 / 0.253 / 0.227 / 0.246 for STPEV against the published 0.239 / 0.253 / 0.227 / 0.246), which confirms the paper’s own point in the strongest form: STPEV is same-company history and has *no definition* under a ticker-disjoint evaluation, so the best model on DEC is one that cannot be

Table 8: Reproduced prior model under leakage-proof evaluation (Result Table 6R, first entry). The HTML head reproduces the paper’s ordering against price baselines at $\tau=3$ (and beats the published number) but not the published $\tau=15$ and $\tau=30$ values, which lie *below the persistence floor* of the shipped labels themselves; at $\tau=30$ every condition is worse than persistence. Moving the same head and labels from the published split to a ticker-disjoint split raises test MSE by 9% / 23% / 46% / 36%; the embargoed temporal split does not.

run in the condition our controls require. The remaining three runnable repositories (KeFVP, DialogueGAT, Sawhney et al. 2020) carry reason codes in Table 82 (Appendix ??): each needs a multi-day environment or corpus rebuild (torch 1.12 and an uninstallable requirements file; a dialogue corpus that must be reconstructed; TF 2.1/Keras 2.3.1), and we did not port them. Of seventeen papers in the lineage, two ship code that runs as released, six ship code with a broken environment or missing data, and nine ship nothing usable.

6.8 Audio by bitrate: the Stage-3 ladder on Earnings25

Earnings25 audio is webcast MP3 at 64 kbps for 281 calls and at 24 kbps or below for 233 (§3.2), so it cannot test “clean” audio; it can test whether the FinCall audio result depends on bitrate *within one quarter*, with the same extractors, heads and controls as Table 5. A single-quarter corpus spans about 105 sessions, so the embargoed temporal split degenerates (five training calls); Table 9 therefore uses the ticker-disjoint split, pooled and per stratum. Audio-plus-past-vol heads beat persistence on Δv (pooled R_{OOS}^2 0.44 at $\tau=3$), but a model given a *random other call’s* audio keeps most of that (0.35): the audio increment is 0.09 at $\tau=3$ and at most 0.05 beyond, and per stratum it is small and sign-inconsistent (64 kbps: 0.03 / 0.00 / 0.08 / -0.02; ≤ 24 kbps: -0.04 / 0.02 / 0.07 / 0.04). With 39 and 55 test calls per stratum these increments are not distinguishable from zero, and the higher-bitrate stratum is not the better one. The FinCall result therefore does not look like an artefact of 40 kbps transcodes: at 64 kbps, on 2025 calls, with a post-cutoff encoder stack, audio is as inert beyond past volatility as it was on FinCall.

Stratum	calls (test)		$\tau=30$
All (64 + ≤ 24 kbps)	476 (95)	real	0.443
		global shuffle	0.351
		increment	0.093
64 kbps	281 (39)	real	0.154
		global shuffle	0.121
		increment	0.033
≤ 24 kbps	233 (55)	real	0.101
		global shuffle	0.141
		increment	-0.040

Table 9: Audio on Earnings25 by bitrate stratum (T9.4: R^2_{OOS} vs. persistence, Δv , ridge on WavLM + past-vol, ticker-disjoint test). The within-ticker shuffle is the identity here (one call per company), so the global shuffle is the control. The audio increment never exceeds 0.09 and has no consistent sign across strata; the 64 kbps stratum is not better than the ≤ 24 kbps one. Full grid (eGeMAPS, emotion2vec+, fusion, MLP heads, all targets); result_table_3_earnings25.csv.

6.9 Post-cutoff lookahead

[pending: Phase 7: the frozen-pipeline evaluation on Earnings25’s post-cutoff calls, no retraining, rule fixed in advance.]

7 Discussion

What the evidence shows. On the completed ladder (Stages 0–4), four statements hold. (i) Strong econometric baselines are hard to beat, and the headline metric (R^2_{OOS} vs. persistence) exposes a long-horizon regime failure (FinCall temporal $\tau=30$) that raw MSE would obscure. (ii) Frozen text content does not cleanly add signal over past-volatility and identity baselines on the identity-robust Δv target, and the reason is measurable: call embeddings encode ticker identity (89.5% probe accuracy on FinCall), and within-ticker transcript shuffles leave performance essentially unchanged—an in-corpus replication of the “Same Company, Same Signal” effect (Yu et al., 2024). (iii) Frozen audio content is inert: audio-only heads sit at R^2_{OOS} zero everywhere, the apparent gains of past-vol-augmented audio heads survive a global audio shuffle, and WavLM embeddings still encode the speaker’s company at 76.3% probe accuracy—identity without predictive content. (iv) Fusion does not rescue the modalities it fuses: multimodal never significantly beats the best unimodal model (RQ2 negative). Together these form one coherent finding: *in this task family, what looks like content signal is overwhelmingly company identity plus past volatility.*

Why this is a benchmark paper. Every one of those statements depends on apparatus the shared substrate does not have. The identity probe and the shuffle controls need a ticker per call, which FinCall-Surprise does

not ship; the ticker-disjoint rows need a split the lineage never reports; the regime finding needs per-year breakdowns over recomputed targets rather than inherited label files; and the embargo makes the temporal numbers honest at all (§3.1). A field that evaluates on files with a zero-day embargo, majority company overlap, and an unstated horizon convention cannot distinguish a model from a memory of which company it is looking at. EC VOL-bench is the repair, and the negative results above are what the repaired substrate reports for the modern representations we could run. The reproduction study (§6.7) is the same test applied to the field’s own models on the field’s own data; it is the part of the argument this draft still owes.

Engaging the strongest positive claims. The strongest recent positive claim—up to 43.8% of 30-day realised-volatility variance from text+acoustic features (Chen et al., 2025)—reports no identity controls, no released code, and a proprietary corpus. Prosody was the modality most plausibly carrying content orthogonal to identity; our modern audio ladder (WavLM, emotion2vec+, eGeMAPS) finds it inert once past volatility is controlled—albeit on the corpus’s 40 kbps source audio (see Limitations and §6.8)—and our shuffle controls predict such gains would not survive a within-ticker (let alone global) shuffle, a check the released suite makes cheap. Likewise, ECC Analyzer (Cao et al., 2024a) reports large MSE reductions from LLM-extracted semantics with closed-API components and no released code; it cannot be reproduced, and we say so in the code-availability audit rather than argue with it.

The pre-registered framing gate. The study was pre-registered with a framing gate: a “positive showcase” spine was to be adopted only if content-bearing models beat the Stage-0/1 floor on Δv or HAR-residual with DM-significant improvements that survived the identity-control suite on at least two of four horizons. The criterion was tested and missed at Stage 2 (text), Stage 3 (audio), and Stage 4 (fusion), and the gate was recorded as resolved toward re-examination in the project’s decision log before the benchmark framing was adopted. We report this so that the framing of this paper can be audited against what the data said and when.

What a clean answer would need. Independent of any model, two facts about the corpora limit every result in this literature and are addressed only by new data: the uniform 40 kbps transcode underlying all FinCall-Surprise acoustics, and the absence of any measured call time of day. The first is why the Earnings25 clean-audio re-run (§6.8) is the single most informative pending experiment—it either makes the audio-inert finding near-unassailable or overturns it; the second is why the timestamp retrofit (§3.4) is part of the benchmark rather than a footnote. **[TODO: Finalise this section once Phase 6R, T9.1, T9.4, and Phase 7 land.]**

8 Conclusion

We audited the evaluation substrate that a decade of multimodal earnings-call volatility prediction shares— inherited label files with a zero-day embargo against 30-day targets, majority company overlap between training and test, no reported ticker-disjoint condition, degenerate label cells, and a horizon convention that downstream work misattributes—and released ECVOL-bench to replace it: reconstructed identity and dates for the one corpus with open audio, targets under both horizon conventions, embargoed and ticker-disjoint splits, strong econometric baselines, and an identity-control suite, all regenerable from one command line. Run over modern open-weight text and audio representations, the benchmark’s answer is consistent and negative: frozen text content reads company identity rather than call content (89.5% ticker-probe accuracy; within ticker shuffles leave performance unchanged); frozen audio content is inert beyond past volatility (its apparent gains survive a global audio shuffle); and fusion never significantly beats the best unimodal model. What the leaderboard literature measures as content signal is, under honest controls, overwhelmingly company identity plus past volatility. *[pending: The reproduction study, the calendar-day recomputation, the clean-audio re-run, and the post-cutoff lookahead complete the argument.]* **[TODO: Finalise the verdict once the pending results land.]**

Limitations

Pending results. The substrate-audit numbers in §3.1 were established in a one-off audit and await regeneration by `ecvol audit substrate`; the reproduction study (§6.7), the calendar-day recomputation (§6.6), the clean-audio re-run (§6.8), and the post-cutoff lookahead (§6.9) are owed. Until the lookahead lands, results on 2019–2021 calls from models with later training cutoffs carry contamination risk by default. **Reproductions run on their own data.** Each prior model is re-evaluated on its original dataset, not on ECVOL-bench to keep “the model fails” separable from “the model does not transfer”; two reproductions carry known costs (KeFVP’s MAEC embeddings are unavailable upstream; DialogueGAT stops at $\tau=15$ and requires a corpus rebuild). **MLP-head instability.** The shallow-MLP heads are seed-unstable (across-seed R_{OOS}^2 std up to ≈ 3.4), which is why the consolidated grid uses deterministic ridge/structural canonicals; MLP cells in the appendix should be read as seed means with that caveat. **The past-vol covariate block is not HAR.** The learned $[v_{\text{pre}}, \text{RV}_{1/5/22}]$ ridge used inside Stage-2–4 heads overfits the COVID regime shift on the temporal split ($R_{\text{OOS}}^2 - 2.0$ at $\tau=15$) where the rigid OLS HAR RV stays robust ($+0.21$); the two are labelled distinctly throughout, and the gap is itself a small finding about structural vs. unconstrained conditioning under regime

shift. **Short-horizon R_{OOS}^2 inflation.** The persistence denominator is noisiest at $\tau=3$ (its MSE is $\sim 2.8\times$ the $\tau=7$ one), so short-horizon R_{OOS}^2 gains are easier to show and should be read accordingly. **Timestamps.** No call in either corpus carries a measured time of day, so every target rests on a flagged after-hours rule; misdated calls add noise to short-horizon targets, and the sensitivity of the results to that rule is unmeasured until the retrofit of §3.4 lands. **Regime sensitivity.** FinCall-Surprise spans COVID; the temporal $\tau=30$ baseline failure shows long-horizon level forecasts are regime-sensitive, which is why we emphasise the identity-robust Δv target and report per-year breakdowns. **MAEC coverage.** The 75.4% price-join rate (2015–2018 delistings) makes MAEC a secondary, comparability corpus; it also ships no raw audio, so the audio ladder is FinCall-only. **Source audio quality.** Every FinCall audio file is a uniform 40 kbps MP3 transcode (eight source sample rates; 14 calls at telephone-grade 8 kHz), so aggressive lossy compression underlies all acoustic features. The audio-inert finding is therefore established on compression-damaged audio and may not transfer to studio-quality recordings—a caveat that applies equally to prior positive claims built on the same corpora, and the reason for the Earnings25 re-run. **No diarisation in v1.** Audio uses per-call pooling, not speaker-resolved features; the gender analysis correspondingly uses a dominant-speaker F0 proxy. **Identity reconstruction is imperfect.** 7.1% of FinCall-Surprise calls (4.7% of earnings calls) remain unresolved and are excluded with a reason code; the released override table records every manual decision.

Ethics Statement

Data and licensing. We use only openly licensed corpora (FinCall-Surprise, Apache-2.0; MAEC, CC-BY-SA-4.0) and open-weight models. Derived numeric features inherit and explicitly state their upstream license; raw audio/transcripts from unclear-license sources are never redistributed (*scripts-not-data*). **Gender confound.** Audio features encode speaker gender and female executives are underrepresented in these corpora, which can produce error disparities (Wang et al., 2024; Lin et al., 2024; Mathur et al., 2022); we analyse and report a gender-confound diagnostic rather than silently propagate the bias. **Intended use.** This is a forecasting-research benchmark, not investment advice; predictions are noisy and regime-sensitive, and should not be used for automated trading. **Reproducibility as an ethical commitment.** Given the documented reproducibility crisis in financial NLP (Tatarinov et al., 2025), we pin the environment, checksum every data file, and regenerate all result tables byte-identically from committed artifacts.

References

- BAAI. BGE-M3 sentence embeddings. Hugging Face (MIT), 2024. URL <https://huggingface.co/BAAI/bge-m3>.
- Mostapha Benhenda. Look-ahead-bench: a standardized benchmark of look-ahead bias in point-in-time LLMs for finance. arXiv:2601.13770, 2026. URL <https://arxiv.org/abs/2601.13770>.
- Yupeng Cao et al. ECC analyzer: Extracting trading signal from earnings conference calls using large language models for stock volatility prediction. In *Proceedings of the 5th ACM International Conference on AI in Finance (ICAIF)*, 2024a. URL <https://arxiv.org/abs/2404.18470>.
- Yupeng Cao et al. RiskLabs: Predicting financial risk using large language model based on multimodal and multi-sources data. arXiv:2404.07452, 2024b. URL <https://arxiv.org/abs/2404.07452>.
- Yupeng Cao et al. FinAudio: A benchmark for audio large language models in financial applications. arXiv:2503.20990, 2025. URL <https://arxiv.org/abs/2503.20990>.
- Chen et al. Distilling numeral information for volatility forecasting. In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM)*, 2021a. doi: 10.1145/3459637.3482089.
- Sanyuan Chen et al. WavLM: Large-scale self-supervised pre-training for full stack speech processing. Hugging Face (MIT): microsoft/wavlm-large, 2021b. URL <https://huggingface.co/microsoft/wavlm-large>.
- Xiaoliang Chen et al. The sound of risk: A multimodal physics-informed acoustic model for forecasting market volatility and enhancing market interpretability. arXiv:2508.18653, 2025. URL <https://arxiv.org/abs/2508.18653>.
- Umberto Collodel. Interpreting the interpreter: Can we model post-ECB conferences volatility with LLM agents? arXiv:2508.13635, 2025. URL <https://arxiv.org/abs/2508.13635>.
- Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 2009. doi: 10.1093/jjfinc/ebp001.
- Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 1995. doi: 10.1080/07350015.1995.10524599.
- Florian Eyben, Martin Wöllmer, and Björn Schuller. openSMILE / eGeMAPS acoustic feature extraction. github.com/audeering/opensmile-python, 2010. URL <https://github.com/audeering/opensmile-python>.
- Zhenyu Gao, Wenxi Jiang, and Yutong Yan. Detecting lookahead bias in LLM forecasts. arXiv:2512.23847, 2025. URL <https://arxiv.org/abs/2512.23847>.
- Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 2020. doi: 10.1093/rfs/hhaa009.
- Songrun He, Linying Lv, Asaf Manela, and Jimmy Wu. Chronologically consistent large language models. arXiv:2502.21206, 2025. URL <https://arxiv.org/abs/2502.21206>.
- Jiazheng Li, Linyi Yang, Barry Smyth, and Ruihai Dong. MAEC: A multimodal aligned earnings conference call dataset for financial risk prediction. In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM)*, 2020. doi: 10.1145/3340531.3412879.
- Yi-Cheng Lin et al. Emo-bias: A large scale evaluation of social bias on speech emotion recognition. arXiv:2406.05065, 2024. URL <https://arxiv.org/abs/2406.05065>.
- Y. Liu et al. AT-FinGPT: Financial risk prediction via an audio-text large language model. *Finance Research Letters*, 2025. URL <https://www.sciencedirect.com/science/article/abs/pii/S1544612325002314>.
- Alejandro Lopez-Lira, Yuehua Tang, and Mingyin Zhu. The memorization problem: Can we trust LLMs’ economic forecasts? arXiv:2504.14765, 2025. URL <https://arxiv.org/abs/2504.14765>.
- Ziyang Ma et al. emotion2vec: Self-supervised pre-training for speech emotion representation. Hugging Face: emotion2vec/emotion2vec_plus_large, 2023. URL https://huggingface.co/emotion2vec/emotion2vec_plus_large.
- Puneet Mathur et al. DocFin: Multimodal financial prediction and bias mitigation using semi-structured documents. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022. URL <https://aclanthology.org/2022.findings-emnlp.139/>.
- Hao Niu et al. KeFVP: Knowledge-enhanced financial volatility prediction. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. URL <https://aclanthology.org/2023.findings-emnlp.770/>.
- ProsusAI. FinBERT: Financial sentiment analysis with bert. Hugging Face, 2019. URL <https://huggingface.co/ProsusAI/finbert>.

Yu Qin and Yi Yang. What you say and how you say it matters: Predicting stock volatility using verbal and vocal cues. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019. URL <https://aclanthology.org/P19-1038/>.

Anisha Roy and Dip Roy. MemGuard-Alpha: Detecting and filtering memorization-contaminated signals in LLM-based financial forecasting via membership inference and cross-model disagreement. arXiv:2603.26797, 2026. URL <https://arxiv.org/abs/2603.26797>.

Yunxin Sang and Yang Bao. DialogueGAT: A graph attention network for financial risk prediction by modeling the dialogues in earnings conference calls. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022. URL <https://aclanthology.org/2022.findings-emnlp.117/>.

Ramit Sawhney, Piyush Khanna, Arshiya Aggarwal, Taru Jain, Puneet Mathur, and Rajiv Ratn Shah. VolTAGE: Volatility forecasting via text audio fusion with graph convolution networks for earnings calls. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020. URL <https://aclanthology.org/2020.emnlp-main.643/>.

Shi et al. Enhancing volatility forecasting in financial markets: A general numeral attachment dataset for understanding earnings calls. In *Proceedings of IJCNLP-AACL 2023*, 2023. URL <https://aclanthology.org/2023.ijcnlp-short.5/>.

Dong Shu, Yanguang Liu, Huopu Zhang, and Mengnan Du. FinCall-Surprise: A large scale multi-modal benchmark for earning surprise prediction. arXiv:2510.03965; data: github.com/Tizzzy/FinCall-Surprise (Apache-2.0), 2025. URL <https://arxiv.org/abs/2510.03965>.

Nikita Tatarinov et al. Language modeling for the future of finance: A survey into metrics, tasks, and data opportunities. arXiv:2504.07274, 2025. URL <https://arxiv.org/abs/2504.07274>.

Wang et al. AMA-LSTM: Pioneering robust and fair financial audio analysis for stock volatility prediction. NAACL 2024 Industry Track; arXiv:2407.18324, 2024. URL <https://arxiv.org/abs/2407.18324>.

Yutong Yan et al. DatedGPT: Preventing lookahead bias in large language models with time-aware pretraining. arXiv:2603.11838, 2026. URL <https://arxiv.org/abs/2603.11838>.

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000
EWMA	0.387*	0.362*	0.145*	-0.219*
HAR-RV	0.464*	0.410*	0.207*	-0.287*
GARCH(1,1)	0.265*	0.193*	-0.140*	-0.541*
GBDT (ticker FE)	0.354*	0.262*	-0.040	-0.839*

Table 10: fincall — temporal split — level- v — R^2_{OOS} vs persistence (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000
EWMA	0.357*	0.199*	-0.009	0.046
HAR-RV	0.495*	0.357*	0.211*	0.229*
GARCH(1,1)	0.356*	0.229*	0.088	0.209*
GBDT (ticker FE)	0.412*	0.278*	0.078	0.116*

Table 11: fincall — ticker_disjoint split — level- v — R^2_{OOS} vs persistence (test) (n=484).

Linyi Yang, Tin Lok James Ng, Barry Smyth, and Ruihai Dong. HTML: Hierarchical transformer-based multi-task learning for volatility prediction. In *Proceedings of The Web Conference (WWW)*, 2020.

Linyi Yang et al. NumHTML: Numeric-oriented hierarchical transformer model for multi-task financial forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2022. URL <https://arxiv.org/abs/2201.01770>.

Ye, Qin, and Xu. Financial risk prediction with multi-round q&a attention network. In *Proceedings of the 29th International Joint Conference on Artificial Intelligence (IJCAI)*, 2020. URL <https://www.ijcai.org/proceedings/2020/631>.

Ding Yu et al. Same company, same signal: The role of identity in earnings call transcripts. arXiv:2412.18029, 2024. URL <https://arxiv.org/abs/2412.18029>.

A Complete Results

The tables below are generated byte-identically by ecvol report from the committed result CSVs (data/results/) and are the source of truth for the curated tables in §6; the curated tables copy selected cells from them. They span both corpora (FinCall-Surprise, MAEC), the level- v and Δv targets, both the R^2_{OOS} and MSE metrics, and the temporal and ticker-disjoint splits at all horizons, under the trading-day convention [pending: calendar-day columns, T9.1].

A.1 Stage-0/1 baselines (Result Table 1)

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	0.000	0.000	0.000	0.000
EWMA	0.387*	0.362*	0.145*	-0.219*	EWMA	0.398*	0.297*	0.119*	-0.144*
HAR-RV	0.464*	0.410*	0.207*	-0.287*	HAR-RV	0.474*	0.379*	0.311*	0.225*
GARCH(1,1)	0.265*	0.193*	-0.140*	-0.541*	GARCH(1,1)	0.306*	0.213*	0.009	-0.321*
GBDT (ticker FE)	0.344*	0.248*	-0.057	-0.904*	GBDT (ticker FE)	0.394*	0.285*	0.227*	0.233*

Table 12: fincall — temporal split — Δv — R^2_{OOS} vs persistence (test) (n=481).

Table 17: maec — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) (n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	1.174	0.419	0.208	0.111
EWMA	0.357*	0.199*	-0.009	0.046	EWMA	0.720*	0.267*	0.178*	0.136*
HAR-RV	0.495*	0.357*	0.211*	0.229*	HAR-RV	0.630*	0.247*	0.165*	0.143*
GARCH(1,1)	0.356*	0.229*	0.088	0.209*	GARCH(1,1)	0.860*	0.335*	0.233*	0.167*
GBDT (ticker FE)	0.407*	0.275*	0.089	0.108	GBDT (ticker FE)	0.758*	0.309*	0.216	0.205*

Table 13: fincall — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) (n=484).

Table 18: fincall — temporal split — level-v — MSE (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	1.142	0.386	0.214	0.233
EWMA	0.444*	0.301*	0.120*	-0.144*	EWMA	0.735*	0.309*	0.216	0.222
HAR-RV	0.507*	0.392*	0.306*	0.206*	HAR-RV	0.577*	0.248*	0.169*	0.180*
GARCH(1,1)	0.387*	0.281*	0.054	-0.309*	GARCH(1,1)	0.735*	0.297*	0.195	0.184*
GBDT (ticker FE)	0.397*	0.333*	0.217*	0.131*	GBDT (ticker FE)	0.672*	0.278*	0.197	0.206*

Table 14: maec — temporal split — level-v — R^2_{OOS} vs persistence (test) (n=507).

Table 19: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	1.174	0.419	0.208	0.111
EWMA	0.398*	0.297*	0.119*	-0.144*	EWMA	0.720*	0.267*	0.178*	0.136*
HAR-RV	0.474*	0.379*	0.311*	0.225*	HAR-RV	0.630*	0.247*	0.165*	0.143*
GARCH(1,1)	0.306*	0.213*	0.009	-0.321*	GARCH(1,1)	0.860*	0.335*	0.233*	0.167*
GBDT (ticker FE)	0.395*	0.306*	0.236*	0.221*	GBDT (ticker FE)	0.771*	0.315*	0.220	0.212*

Table 15: maec — ticker_disjoint split — level-v — R^2_{OOS} vs persistence (test) (n=514).

Table 20: fincall — temporal split — Δv — MSE (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	1.142	0.386	0.214	0.233
EWMA	0.444*	0.301*	0.120*	-0.144*	EWMA	0.735*	0.309*	0.216	0.222
HAR-RV	0.507*	0.392*	0.306*	0.206*	HAR-RV	0.577*	0.248*	0.169*	0.180*
GARCH(1,1)	0.387*	0.281*	0.054	-0.309*	GARCH(1,1)	0.735*	0.297*	0.195	0.184*
GBDT (ticker FE)	0.407*	0.340*	0.213*	0.186*	GBDT (ticker FE)	0.678*	0.279*	0.195	0.208

Table 16: maec — temporal split — Δv — R^2_{OOS} vs persistence (test) (n=507).

Table 21: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.383	0.571	0.318	0.189	Persistence	0.000	0.000	0.000	0.000
EWMA	0.768*	0.399*	0.280*	0.216*	EWMA	0.318*	0.292*	0.079	-0.071
HAR-RV	0.682*	0.347*	0.221*	0.150*	HAR-RV	0.476*	0.426*	0.255*	0.179*
GARCH(1,1)	0.848*	0.411*	0.301	0.247*	GARCH(1,1)	0.350*	0.299*	0.126	0.088
GBDT (ticker FE)	0.834*	0.381*	0.249*	0.164*	GBDT (ticker FE)	0.368*	0.333*	0.117*	0.076

Table 22: maec — temporal split — level-v — MSE (test) (n=507).

Table 27: fincall — ticker_disjoint split — level-v — R^2_{OOS} vs persistence (test) — calendar-day τ (n=484).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.268	0.492	0.302	0.184	Persistence	0.000	0.000	0.000	0.000
EWMA	0.764*	0.346*	0.266*	0.210*	EWMA	0.247*	0.398*	0.256*	-0.006
HAR-RV	0.667*	0.306*	0.208*	0.142*	HAR-RV	0.380*	0.448*	0.323*	0.017
GARCH(1,1)	0.881*	0.384*	0.296	0.242*	GARCH(1,1)	0.161*	0.257*	0.036	-0.341*
GBDT (ticker FE)	0.767*	0.342*	0.230*	0.143*	GBDT (ticker FE)	0.224*	0.307*	0.101	-0.251*

Table 23: maec — ticker_disjoint split — level-v — MSE (test) (n=514).

Table 28: fincall — temporal split — Δv — R^2_{OOS} vs persistence (test) — calendar-day τ (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.383	0.571	0.318	0.189	Persistence	0.000	0.000	0.000	0.000
EWMA	0.768*	0.399*	0.280*	0.216*	EWMA	0.318*	0.292*	0.079	-0.071
HAR-RV	0.682*	0.347*	0.221*	0.150*	HAR-RV	0.476*	0.426*	0.255*	0.179*
GARCH(1,1)	0.848*	0.411*	0.301	0.247*	GARCH(1,1)	0.350*	0.299*	0.126	0.088
GBDT (ticker FE)	0.820*	0.377*	0.250*	0.154*	GBDT (ticker FE)	0.350*	0.339*	0.130*	0.076

Table 24: maec — temporal split — Δv — MSE (test) (n=507).

Table 29: fincall — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) — calendar-day τ (n=484).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.268	0.492	0.302	0.184	Persistence	0.000	0.000	0.000	0.000
EWMA	0.764*	0.346*	0.266*	0.210*	EWMA	0.510*	0.342*	0.230*	-0.014
HAR-RV	0.667*	0.306*	0.208*	0.142*	HAR-RV	0.574*	0.419*	0.354*	0.235*
GARCH(1,1)	0.881*	0.384*	0.296	0.242*	GARCH(1,1)	0.442*	0.316*	0.187*	-0.145
GBDT (ticker FE)	0.769*	0.352*	0.233*	0.141*	GBDT (ticker FE)	0.472*	0.343*	0.303*	0.156*

Table 25: maec — ticker_disjoint split — Δv — MSE (test) (n=514).

Table 30: maec — temporal split — level-v — R^2_{OOS} vs persistence (test) — calendar-day τ (n=507).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	0.000	0.000	0.000	0.000
EWMA	0.247*	0.398*	0.256*	-0.006	EWMA	0.526*	0.335*	0.205*	-0.034
HAR-RV	0.380*	0.448*	0.323*	0.017	HAR-RV	0.601*	0.404*	0.345*	0.246*
GARCH(1,1)	0.161*	0.257*	0.036	-0.341*	GARCH(1,1)	0.475*	0.249*	0.099	-0.178
GBDT (ticker FE)	0.241*	0.311*	0.120*	-0.216*	GBDT (ticker FE)	0.538*	0.311*	0.298*	0.194*

Table 26: fincall — temporal split — level-v — R^2_{OOS} vs persistence (test) — calendar-day τ (n=481).

Table 31: maec — ticker_disjoint split — level-v — R^2_{OOS} vs persistence (test) — calendar-day τ (n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	2.182	0.580	0.258	0.195
EWMA	0.510*	0.342*	0.230*	-0.014	EWMA	1.488*	0.411*	0.238	0.209
HAR-RV	0.574*	0.419*	0.354*	0.235*	HAR-RV	1.144*	0.333*	0.192*	0.160*
GARCH(1,1)	0.442*	0.316*	0.187*	-0.145	GARCH(1,1)	1.419*	0.407*	0.226	0.178
GBDT (ticker FE)	0.443*	0.348*	0.312*	0.177*	GBDT (ticker FE)	1.418*	0.383*	0.225*	0.180

Table 32: maec — temporal split — Δv — R^2_{OOS} vs persistence (test) — calendar-day τ (n=507).

Table 37: fincall — ticker_disjoint split — Δv — MSE (test) — calendar-day τ (n=484).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	2.391	0.780	0.407	0.236
EWMA	0.526*	0.335*	0.205*	-0.034	EWMA	1.172*	0.513*	0.314*	0.239
HAR-RV	0.601*	0.404*	0.345*	0.246*	HAR-RV	1.018*	0.453*	0.263*	0.181*
GARCH(1,1)	0.475*	0.249*	0.099	-0.178	GARCH(1,1)	1.335*	0.533*	0.331*	0.270
GBDT (ticker FE)	0.495*	0.321*	0.295*	0.202*	GBDT (ticker FE)	1.262*	0.512*	0.284*	0.199*

Table 33: maec — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) — calendar-day τ (n=514).

Table 38: maec — temporal split — level-v — MSE (test) — calendar-day τ (n=507).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	2.307	0.597	0.288	0.153	Persistence	2.664	0.720	0.363	0.228
EWMA	1.738*	0.359*	0.214*	0.154	EWMA	1.262*	0.479*	0.288*	0.235
HAR-RV	1.431*	0.329*	0.195*	0.150	HAR-RV	1.064*	0.429*	0.238*	0.172*
GARCH(1,1)	1.934*	0.441*	0.275	0.202*	GARCH(1,1)	1.399*	0.537*	0.324	0.267
GBDT (ticker FE)	1.752*	0.411*	0.254*	0.186*	GBDT (ticker FE)	1.230*	0.496*	0.255*	0.184*

Table 34: fincall — temporal split — level-v — MSE (test) — calendar-day τ (n=481).

Table 39: maec — ticker_disjoint split — level-v — MSE (test) — calendar-day τ (n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	2.182	0.580	0.258	0.195	Persistence	2.391	0.780	0.407	0.236
EWMA	1.488*	0.411*	0.238	0.209	EWMA	1.172*	0.513*	0.314*	0.239
HAR-RV	1.144*	0.333*	0.192*	0.160*	HAR-RV	1.018*	0.453*	0.263*	0.181*
GARCH(1,1)	1.419*	0.407*	0.226	0.178	GARCH(1,1)	1.335*	0.533*	0.331*	0.270
GBDT (ticker FE)	1.379*	0.387*	0.228*	0.180	GBDT (ticker FE)	1.331*	0.508*	0.280*	0.194*

Table 35: fincall — ticker_disjoint split — level-v — MSE (test) — calendar-day τ (n=484).

Table 40: maec — temporal split — Δv — MSE (test) — calendar-day τ (n=507).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	2.307	0.597	0.288	0.153	Persistence	2.664	0.720	0.363	0.228
EWMA	1.738*	0.359*	0.214*	0.154	EWMA	1.262*	0.479*	0.288*	0.235
HAR-RV	1.431*	0.329*	0.195*	0.150	HAR-RV	1.064*	0.429*	0.238*	0.172*
GARCH(1,1)	1.934*	0.441*	0.275	0.202*	GARCH(1,1)	1.399*	0.537*	0.324	0.267
GBDT (ticker FE)	1.791*	0.414*	0.259	0.192*	GBDT (ticker FE)	1.345*	0.489*	0.256*	0.182*

Table 36: fincall — temporal split — Δv — MSE (test) — calendar-day τ (n=481).

Table 41: maec — ticker_disjoint split — Δv — MSE (test) — calendar-day τ (n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.439*	0.319	0.037	-0.577	Ridge (text)	0.028*	0.030*	0.015	-0.144*
MLP (text)	0.147*	-0.241*	-1.101*	-2.633*	MLP (text)	-0.108*	-0.175*	-0.265*	-1.042
Ridge (past-vol)	0.415*	-1.052	-2.025	-0.371*	Ridge (past-vol)	0.415*	-1.053	-2.858	-0.667
MLP (past-vol)	0.456*	-0.228	-2.706	-2.477	MLP (past-vol)	0.390	-1.017	-2.249	-0.760
Ridge (text+vol)	0.448*	0.328	0.045	-1.091	Ridge (text+vol)	-1.678	-3.076	-3.586	-0.468
MLP (text+vol)	-2.092	-7.189	-18.384	-26.901	MLP (text+vol)	-1.186	-1.571	-2.112	-1.295

Table 42: fincall — temporal split — level-v — R^2_{OOS} vs persistence (test) (n=481).

Table 44: fincall — temporal split — Δv — R^2_{OOS} vs persistence (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.416	0.194*	-0.047*	0.112	Ridge (text)	-0.044*	-0.026*	0.008	0.171
MLP (text)	-0.362*	-1.801*	-3.485*	-2.872*	MLP (text)	-0.195*	-0.271*	-0.339*	-0.204*
Ridge (past-vol)	0.432	0.306	0.191*	0.224*	Ridge (past-vol)	0.451	0.306	0.191*	0.224*
MLP (past-vol)	0.482*	0.365*	0.243*	0.212*	MLP (past-vol)	0.492*	0.363*	0.230*	0.249*
Ridge (text+vol)	0.418	0.211	-0.006	0.139	Ridge (text+vol)	0.223*	0.129*	0.094	0.236
MLP (text+vol)	-0.254*	-1.556*	-3.005*	-2.486*	MLP (text+vol)	0.275*	0.140*	-0.061*	0.051

Table 43: fincall — ticker_disjoint split — level-v — R^2_{OOS} vs persistence (test) (n=484).

Table 45: fincall — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) (n=484).

A.2 Stage-2 text heads (Result Table 2)

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.478*	0.225*	-0.033*	-0.322*
MLP (text)	-0.748*	-1.753*	-3.272*	-6.016*
Ridge (past-vol)	0.437	0.222*	0.168	0.208
MLP (past-vol)	0.472*	0.375	0.301*	0.247*
Ridge (text+vol)	0.482*	0.279	0.132	-0.013*
MLP (text+vol)	-0.405*	-1.460*	-2.977*	-5.058*

Table 46: maec — temporal split — level-v — R^2_{OOS} vs persistence (test) (n=507).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.466*	0.277	0.117*	-0.036*
MLP (text)	-0.115*	-0.956*	-1.583*	-2.635*
Ridge (past-vol)	0.455*	0.329	0.261	0.262
MLP (past-vol)	0.482*	0.369*	0.281	0.250
Ridge (text+vol)	0.451	0.299	0.186	0.090*
MLP (text+vol)	-0.077*	-0.884*	-1.418*	-2.992*

Table 47: maec — ticker_disjoint split — level-v — R^2_{OOS} vs persistence (test) (n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.017*	0.013*	0.047*	-0.074*
MLP (text)	-0.301*	-0.164*	-0.338*	-0.511*
Ridge (past-vol)	0.455	0.313	0.244	0.212
MLP (past-vol)	0.497*	0.395*	0.323*	0.261*
Ridge (text+vol)	-0.005*	-0.142*	0.211	0.138
MLP (text+vol)	0.259*	0.242*	0.068*	-0.080*

Table 48: maec — temporal split — Δv — R^2_{OOS} vs persistence (test) (n=507).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.049*	0.056*	0.048*	0.047*
MLP (text)	-0.051*	-0.108*	-0.165*	-0.199*
Ridge (past-vol)	0.455*	0.329	0.261	0.262
MLP (past-vol)	0.489*	0.389*	0.328*	0.294*
Ridge (text+vol)	0.330	0.142*	0.204	0.163
MLP (text+vol)	0.366	0.244	0.120*	-0.111*

Table 49: maec — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) (n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.659*	0.285	0.200	0.176
MLP (text)	1.001*	0.519*	0.437*	0.405*
Ridge (past-vol)	0.687*	0.859	0.629	0.153*
MLP (past-vol)	0.639*	0.514	0.771	0.387
Ridge (text+vol)	0.648*	0.281	0.199	0.233
MLP (text+vol)	3.631	3.428	4.032	3.109

Table 50: fincall — temporal split — level-v — MSE (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.667	0.311*	0.224*	0.207
MLP (text)	1.556*	1.080*	0.958*	0.903*
Ridge (past-vol)	0.649	0.268	0.173*	0.181*
MLP (past-vol)	0.592*	0.245*	0.162*	0.184*
Ridge (text+vol)	0.665	0.304	0.215	0.201
MLP (text+vol)	1.432*	0.986*	0.856*	0.813*

Table 51: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	1.141*	0.406*	0.205	0.127*
MLP (text)	1.302*	0.492*	0.263*	0.228
Ridge (past-vol)	0.687*	0.860	0.803	0.186
MLP (past-vol)	0.717	0.844	0.676	0.196
Ridge (text+vol)	3.145	1.706	0.954	0.164
MLP (text+vol)	2.567	1.076	0.647	0.256

Table 52: fincall — temporal split — Δv — MSE (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	1.193*	0.396*	0.212	0.193
MLP (text)	1.365*	0.490*	0.286*	0.281*
Ridge (past-vol)	0.627	0.268	0.173*	0.181*
MLP (past-vol)	0.580*	0.246*	0.164*	0.175*
Ridge (text+vol)	0.888*	0.336*	0.194	0.178
MLP (text+vol)	0.828*	0.332*	0.227*	0.221

Table 53: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.722*	0.443*	0.329*	0.250*
MLP (text)	2.416*	1.573*	1.359*	1.325*
Ridge (past-vol)	0.778	0.445*	0.265	0.150
MLP (past-vol)	0.730*	0.357	0.222*	0.142*
Ridge (text+vol)	0.716*	0.412	0.276	0.191*
MLP (text+vol)	1.943*	1.405*	1.266*	1.144*

Table 54: maec — temporal split — level-v — MSE (test) (n=507).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	0.677*	0.356	0.266*	0.190*
MLP (text)	1.413*	0.963*	0.779*	0.668*
Ridge (past-vol)	0.691*	0.330	0.223	0.136
MLP (past-vol)	0.656*	0.311*	0.217	0.138
Ridge (text+vol)	0.695	0.345	0.246	0.167*
MLP (text+vol)	1.365*	0.927*	0.729*	0.734*

Table 55: maec — ticker_disjoint split — level-v — MSE (test) (n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	1.358*	0.564*	0.303*	0.203*
MLP (text)	1.798*	0.665*	0.426*	0.285*
Ridge (past-vol)	0.753	0.392	0.241	0.149
MLP (past-vol)	0.695*	0.346*	0.215*	0.140*
Ridge (text+vol)	1.390*	0.653*	0.251	0.163
MLP (text+vol)	1.025*	0.433*	0.297*	0.204*

Table 56: maec — temporal split — Δv — MSE (test) (n=507).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Ridge (text)	1.206*	0.464*	0.287*	0.175*
MLP (text)	1.332*	0.546*	0.351*	0.220*
Ridge (past-vol)	0.691*	0.330	0.223	0.136
MLP (past-vol)	0.648*	0.301*	0.203*	0.130*
Ridge (text+vol)	0.850	0.422*	0.240	0.154
MLP (text+vol)	0.804	0.372	0.265*	0.204*

Table 57: maec — ticker_disjoint split — Δv — MSE (test) (n=514).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
ridge eGeMAPS	0.362	0.126*	-0.338*	-1.109*
mlp eGeMAPS	-9.446	-22.924	-39.453	-63.845
ridge eGeMAPS +vol	0.347	0.258	-0.046	-0.668
mlp eGeMAPS +vol	-9.226	-21.289	-46.203	-77.488
ridge WavLM	0.359	0.122*	-0.427*	-1.167*
mlp WavLM	-0.117*	-0.965*	-2.387*	-5.448*
ridge WavLM +vol	0.343	0.245	-0.390	-1.599
mlp WavLM +vol	-4.487	-14.363	-32.806	-59.317
ridge emotion2vec+	0.353	0.138*	-0.289*	-1.025*
mlp emotion2vec+	-0.385*	-1.880*	-4.314*	-7.784*
ridge emotion2vec+ +vol	0.401	-0.583	-1.139	-0.422*
mlp emotion2vec+ +vol	-6.056	-20.286	-41.247	-64.876
ridge WavLM+eGeMAPS	0.358	0.116*	-0.379*	-1.172*
mlp WavLM+eGeMAPS	-2.476	-6.938	-11.288	-17.515
ridge WavLM+eGeMAPS +vol	0.338	0.221	-0.139	-1.661
mlp WavLM+eGeMAPS +vol	-5.037	-14.919	-30.138	-43.263
ridge WavLM+text +vol	0.443*	0.321	0.014	-1.142
mlp WavLM+text +vol	-2.902	-8.912	-22.030	-33.025

Table 58: fincall — temporal split — level-v — R^2_{OOS} vs persistence (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
ridge eGeMAPS	0.410	0.106*	-0.238*	-0.106*
mlp eGeMAPS	-0.074	-1.156*	-1.969*	-1.835*
ridge eGeMAPS +vol	0.416	0.270	0.164*	0.116*
mlp eGeMAPS +vol	-0.338	-1.497	-2.709	-2.606*
ridge WavLM	0.397	0.092*	-0.235*	-0.092*
mlp WavLM	-0.355*	-1.902*	-4.384*	-4.021*
ridge WavLM +vol	0.403	0.137*	0.088	0.116*
mlp WavLM +vol	-0.320*	-1.388*	-2.985*	-3.161*
ridge emotion2vec+	0.409	0.125*	-0.202*	-0.061*
mlp emotion2vec+	-1.476	-5.431	-10.233	-9.811*
ridge emotion2vec+ +vol	0.416	0.298	0.195*	0.211*
mlp emotion2vec+ +vol	-1.243	-4.109	-8.681	-7.111*
ridge WavLM+eGeMAPS	0.394	0.094*	-0.230*	-0.092*
mlp WavLM+eGeMAPS	-0.308*	-1.393*	-2.760*	-2.331*
ridge WavLM+eGeMAPS +vol	0.400	0.137*	0.073	0.116*
mlp WavLM+eGeMAPS +vol	-0.302*	-1.323*	-2.551*	-2.151*
ridge WavLM+text +vol	0.412	0.205	0.001	0.116*
mlp WavLM+text +vol	-0.136*	-1.278*	-2.540*	-2.041*

Table 59: fincall — ticker_disjoint split — level-v — R^2_{OOS} vs persistence (test) (n=484).

A.3 Stage-3 audio heads (Result Table 3)

FinCall-Surprise only (MAEC ships no raw audio).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
ridge eGeMAPS	-0.000*	-0.007*	-0.015	-0.000*
mlp eGeMAPS	-0.492*	-0.218*	-0.272*	-0.335*
ridge eGeMAPS +vol	0.219	-1.834	-4.099	-0.611*
mlp eGeMAPS +vol	0.187	-1.680	-5.488	-5.311*
ridge WavLM	-0.005*	-0.007*	-0.045	-0.031*
mlp WavLM	-0.176*	-0.248*	-0.449*	-0.911*
ridge WavLM +vol	0.265	-1.026	-2.032	-0.511*
mlp WavLM +vol	-1.219	-2.099	-2.908	-2.411*
ridge emotion2vec+	-0.018*	-0.028*	-0.056	-0.000*
mlp emotion2vec+	-0.096*	-0.256*	-0.468*	-1.241*
ridge emotion2vec+ +vol	0.229	-2.052	-3.031	-0.611*
mlp emotion2vec+ +vol	-1.692	-3.466	-3.611	-2.311*
ridge WavLM+eGeMAPS	-0.005*	0.001*	-0.015	-0.021*
mlp WavLM+eGeMAPS	-0.296*	-0.497*	-0.641*	-1.341*
ridge WavLM+eGeMAPS +vol	0.214	-1.099	-3.706	-0.511*
mlp WavLM+eGeMAPS +vol	-1.810	-3.234	-3.991	-3.111*
ridge WavLM+text +vol	-1.697	-3.043	-0.323	-0.411*
mlp WavLM+text +vol	-1.456	-2.193	-3.917	-3.111*

Table 60: fincall — temporal split — Δv — R^2_{OOS} vs persistence (test) (n=481).

Model	$\tau=3$	$\tau=7$	Model	$\tau=30$	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
ridge eGeMAPS	-0.011*	-0.013*	ridge eGeMAPS	0.002	0.674	0.345*	0.265*	0.258*
mlp eGeMAPS	-0.167*	-0.231*	mlp eGeMAPS	-0.171*	1.227	0.832*	0.634*	0.666*
ridge eGeMAPS +vol	0.415	0.253	ridge eGeMAPS +vol	0.002	0.667	0.281	0.179*	0.188
mlp eGeMAPS +vol	0.353	0.207	mlp eGeMAPS +vol	0.070	1.529	0.963	0.792	0.843
ridge WavLM	-0.038*	-0.035*	ridge WavLM	-0.027*	0.689	0.350*	0.264*	0.256*
mlp WavLM	-0.248*	-0.323*	mlp WavLM	-0.351*	1.547*	1.119*	1.150*	1.172*
ridge WavLM +vol	0.337*	0.171*	ridge WavLM +vol	0.024	0.682	0.333*	0.195	0.198
mlp WavLM +vol	0.160*	0.009*	mlp WavLM +vol	-0.174*	1.508*	0.921*	0.852*	0.971*
ridge emotion2vec+	-0.002*	-0.003*	ridge emotion2vec+	0.002	0.675	0.338*	0.257*	0.249*
mlp emotion2vec+	-0.257*	-0.397*	mlp emotion2vec+	-0.492*	2.828	2.480	2.400	2.539
ridge emotion2vec+ +vol	0.435	0.297	ridge emotion2vec+ +vol	0.108	0.667	0.271	0.172*	0.182*
mlp emotion2vec+ +vol	0.105*	-0.139*	mlp emotion2vec+ +vol	-0.255*	2.562	1.970	2.068	1.909
ridge WavLM+eGeMAPS	-0.050*	-0.042*	ridge WavLM+eGeMAPS	-0.026*	0.692	0.350*	0.263*	0.256*
mlp WavLM+eGeMAPS	-0.225*	-0.366*	mlp WavLM+eGeMAPS	-0.451*	1.493*	0.923*	0.803*	0.778*
ridge WavLM+eGeMAPS +vol	0.304*	0.147*	ridge WavLM+eGeMAPS +vol	0.020	0.686	0.333*	0.198	0.200
mlp WavLM+eGeMAPS +vol	0.109*	-0.004*	mlp WavLM+eGeMAPS +vol	-0.111*	1.487*	0.896*	0.759*	0.736*
ridge WavLM+text +vol	0.159*	0.072*	ridge WavLM+text +vol	0.023	0.671	0.307	0.214	0.199
mlp WavLM+text +vol	0.228*	0.096*	mlp WavLM+text +vol	-0.084*	1.298*	0.878*	0.756*	0.709*

Table 61: fincall — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) (n=484).

Table 63: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Model	$\tau=3$	$\tau=7$	Model	$\tau=30$	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
ridge eGeMAPS	0.750	0.366*	ridge eGeMAPS	0.278*	1.175*	0.421*	0.211	0.112*
mlp eGeMAPS	12.267	10.015	mlp eGeMAPS	8.416	1.752*	0.510*	0.265*	0.149*
ridge eGeMAPS +vol	0.767	0.311	ridge eGeMAPS +vol	0.218	0.917	1.186	1.061	0.182
mlp eGeMAPS +vol	12.008	9.331	mlp eGeMAPS +vol	9.819	0.954	1.122	1.350	0.704
ridge WavLM	0.753	0.368*	ridge WavLM	0.297*	1.180*	0.422*	0.217	0.115*
mlp WavLM	1.312*	0.823*	mlp WavLM	0.704	1.381*	0.523*	0.301*	0.221
ridge WavLM +vol	0.772	0.316	ridge WavLM +vol	0.289	0.863	0.848	0.631	0.174
mlp WavLM +vol	6.443	6.432	mlp WavLM +vol	7.032	2.606	1.297	0.813	0.389
ridge emotion2vec+	0.759	0.361*	ridge emotion2vec+	0.226	1.196*	0.431*	0.220	0.118*
mlp emotion2vec+	1.626*	1.206*	mlp emotion2vec+	0.979	1.288*	0.526*	0.305*	0.250*
ridge emotion2vec+ +vol	0.703	0.663	ridge emotion2vec+ +vol	0.158	0.905	1.278	0.838	0.187
mlp emotion2vec+ +vol	8.286	8.911	mlp emotion2vec+ +vol	7.340	3.161	1.870	0.959	0.370
ridge WavLM+eGeMAPS	0.754	0.370*	ridge WavLM+eGeMAPS	0.287*	1.180*	0.418*	0.211	0.115*
mlp WavLM+eGeMAPS	4.082	3.323	mlp WavLM+eGeMAPS	2.556	1.522*	0.627*	0.341*	0.261*
ridge WavLM+eGeMAPS +vol	0.777	0.326	ridge WavLM+eGeMAPS +vol	0.237	0.923	0.879	0.979	0.172
mlp WavLM+eGeMAPS +vol	7.089	6.664	mlp WavLM+eGeMAPS +vol	6.477	3.300	1.773	1.038	0.466
ridge WavLM+text +vol	0.654*	0.284	ridge WavLM+text +vol	0.205	3.167	1.692	0.275	0.161
mlp WavLM+text +vol	4.582	4.150	mlp WavLM+text +vol	4.791	2.884	1.337	1.023	0.458

Table 62: fincall — temporal split — level-v — MSE (test) (n=481).

Table 64: fincall — temporal split — Δv — MSE (test) (n=481).

Model	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
ridge eGeMAPS	1.155*	0.391*	0.214	0.233
mlp eGeMAPS	1.333*	0.475*	0.251*	0.258*
ridge eGeMAPS +vol	0.668	0.288	0.180	0.186*
mlp eGeMAPS +vol	0.739	0.306	0.199	0.217
ridge WavLM	1.185*	0.399*	0.219*	0.228
mlp WavLM	1.426*	0.510*	0.289*	0.275*
ridge WavLM +vol	0.758*	0.320*	0.190	0.184*
mlp WavLM +vol	0.959*	0.382*	0.251*	0.261*
ridge emotion2vec+	1.144*	0.387*	0.211	0.233
mlp emotion2vec+	1.436*	0.539*	0.319*	0.335*
ridge emotion2vec+ +vol	0.645	0.271	0.171*	0.182*
mlp emotion2vec+ +vol	1.022*	0.439*	0.268*	0.275*
ridge WavLM+eGeMAPS	1.199*	0.402*	0.219*	0.227
mlp WavLM+eGeMAPS	1.400*	0.527*	0.312*	0.316*
ridge WavLM+eGeMAPS +vol	0.795*	0.329*	0.204	0.184*
mlp WavLM+eGeMAPS +vol	1.017*	0.387*	0.239*	0.260*
ridge WavLM+text +vol	0.961*	0.358*	0.207	0.186
mlp WavLM+text +vol	0.882*	0.348*	0.232*	0.216

Table 65: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.464	0.410	0.207	-0.287	HAR-RV	0.464	0.410	0.207	-0.287
GBDT (ticker FE)	0.354	0.262	-0.040	-0.839	GBDT (ticker FE)	0.344	0.248	-0.057	-0.904
Text (BGE+vol)	0.448	0.328	0.045	-1.091	Text (BGE+vol)	-1.678	-3.076	-3.586	-0.468
Audio (WavLM+eG+vol)	0.338	0.221	-0.139	-1.661	Audio (WavLM+eG+vol)	0.214	-1.099	-3.706	-0.548
Fusion (stack)	0.448	0.370	0.125	-1.045	Fusion (stack)	0.232	-0.326	-1.828	-0.010*

Table 66: fincall — temporal split — level-v — R^2_{OOS} vs persistence (test) (n=481).

Table 68: fincall — temporal split — Δv — R^2_{OOS} vs persistence (test) (n=481).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$	Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000	Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.495	0.357	0.211	0.229	HAR-RV	0.495	0.357	0.211	0.229
GBDT (ticker FE)	0.412	0.278	0.078	0.116	GBDT (ticker FE)	0.407	0.275	0.089	0.108
Text (BGE+vol)	0.418	0.211	-0.006	0.132	Text (BGE+vol)	0.223*	0.129*	0.094	0.236
Audio (WavLM+eG+vol)	0.400	0.137	0.073	0.144	Audio (WavLM+eG+vol)	0.304*	0.147*	0.044	0.209
Fusion (stack)	0.450	0.264	0.050	0.171	Fusion (stack)	0.436	0.292	0.133	0.244*

Table 67: fincall — ticker_disjoint split — level-v — R^2_{OOS} vs persistence (test) (n=484).

Table 69: fincall — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) (n=484).

A.4 Consolidated ablation grid (Result Table 4)

One pre-registered canonical model per stage; * marks Holm-corrected significance vs. Stage-1 within the Δv confirmatory family (two-sided—read direction from the value).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.507	0.392	0.306	0.206
GBDT (ticker FE)	0.397	0.333	0.217	0.131
Text (BGE+vol)	0.482	0.279	0.132	-0.013
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 70: maec — temporal split — level-v — R^2_{OOS} vs persistence (test) (n=507).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.474	0.379	0.311	0.225
GBDT (ticker FE)	0.395	0.306	0.236	0.221
Text (BGE+vol)	0.451	0.299	0.186	0.090
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 71: maec — ticker_disjoint split — level-v — R^2_{OOS} vs persistence (test) (n=514).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.507	0.392	0.306	0.206
GBDT (ticker FE)	0.407	0.340	0.213	0.186
Text (BGE+vol)	-0.005*	-0.142*	0.211	0.138
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 72: maec — temporal split — Δv — R^2_{OOS} vs persistence (test) (n=507).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	0.000	0.000	0.000	0.000
HAR-RV	0.474	0.379	0.311	0.225
GBDT (ticker FE)	0.394	0.285	0.227	0.233
Text (BGE+vol)	0.330	0.142*	0.204	0.163
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 73: maec — ticker_disjoint split — Δv — R^2_{OOS} vs persistence (test) (n=514).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.174	0.419	0.208	0.111
HAR-RV	0.630	0.247	0.165	0.143
GBDT (ticker FE)	0.758	0.309	0.216	0.205
Text (BGE+vol)	0.648	0.281	0.199	0.233
Audio (WavLM+eG+vol)	0.777	0.326	0.237	0.296
Fusion (stack)	0.649	0.264	0.182	0.225

Table 74: fincall — temporal split — level-v — MSE (test) (n=481).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.142	0.386	0.214	0.233
HAR-RV	0.577	0.248	0.169	0.180
GBDT (ticker FE)	0.672	0.278	0.197	0.206
Text (BGE+vol)	0.665	0.304	0.215	0.201
Audio (WavLM+eG+vol)	0.686	0.333	0.198	0.200
Fusion (stack)	0.628	0.284	0.203	0.193

Table 75: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.174	0.419	0.208	0.111
HAR-RV	0.630	0.247	0.165	0.143
GBDT (ticker FE)	0.771	0.315	0.220	0.212
Text (BGE+vol)	3.145	1.706	0.954	0.164
Audio (WavLM+eG+vol)	0.923	0.879	0.979	0.172
Fusion (stack)	0.902	0.555	0.588	0.113*

Table 76: fincall — temporal split — Δv — MSE (test) (n=481).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.142	0.386	0.214	0.233
HAR-RV	0.577	0.248	0.169	0.180
GBDT (ticker FE)	0.678	0.279	0.195	0.208
Text (BGE+vol)	0.888*	0.336*	0.194	0.178
Audio (WavLM+eG+vol)	0.795*	0.329*	0.204	0.184
Fusion (stack)	0.644	0.273	0.185	0.176*

Table 77: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.383	0.571	0.318	0.189
HAR-RV	0.682	0.347	0.221	0.150
GBDT (ticker FE)	0.834	0.381	0.249	0.164
Text (BGE+vol)	0.716	0.412	0.276	0.191
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 78: maec — temporal split — level-v — MSE (test) (n=507).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.268	0.492	0.302	0.184
HAR-RV	0.667	0.306	0.208	0.142
GBDT (ticker FE)	0.767	0.342	0.230	0.143
Text (BGE+vol)	0.695	0.345	0.246	0.167
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 79: maec — ticker_disjoint split — level-v — MSE (test) (n=514).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.383	0.571	0.318	0.189
HAR-RV	0.682	0.347	0.221	0.150
GBDT (ticker FE)	0.820	0.377	0.250	0.154
Text (BGE+vol)	1.390*	0.653*	0.251	0.163
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 80: maec — temporal split — Δv — MSE (test) (n=507).

Stage	$\tau=3$	$\tau=7$	$\tau=15$	$\tau=30$
Persistence	1.268	0.492	0.302	0.184
HAR-RV	0.667	0.306	0.208	0.142
GBDT (ticker FE)	0.769	0.352	0.233	0.141
Text (BGE+vol)	0.850	0.422*	0.240	0.154
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 81: maec — ticker_disjoint split — Δv — MSE (test) (n=514).

A.5 Substrate audit (Result Table 5R)

The full per-file audit behind Table 1 (file SHA-256 per repository commit, per-split counts, embargo and duplicate checks, sentinel-zero cells per split, binary-column detection per file, and the per-horizon MAEC label diff) is released as `data/results/result_table_5r.csv` and `result_table_5r_labels.csv`, with the lineage mirror manifest under `raw/ref/lineage/`.

A.6 Reproductions and code-availability audit (Result Table 6R)

The complete reproduction grid behind Table 8 is released as `data/results/result_table_6r.csv` (HTML on EC: three seeds, tuned α , persistence per cell) and `result_table_6r_scss.csv` (Same-Company-Same-Signal on DEC: PEV and STPEV per rolling quarter and horizon, 2021–2023, with the published 2023 cells alongside). Table 82 is the code-availability audit.

Paper	venue	probe (date)	verdict	reason
Qin & Yang, What You Say and How You Say It Matters	ACL 2019	200, 0.0 code files, pushed 2022-08-04 (2026-09-02)	partial	dataset released (Drive, PyTorch model code, no
Yang et al., HTML	WWW 2020	200, 15.0 code files, pushed 2024-01-05 (2026-09-02)	runnable	TF 2.1 / Keras 2.3.1 pin
Sawhney et al., multimodal financial forecasting	ACM MM 2020	200, 112.0 code files, pushed 2020-10-26 (2026-09-02)	partial	ment rebuild)
Sawhney et al., VoltAGE	EMNLP 2020	200, 5.0 code files, pushed 2021-11-16 (2026-09-02)	partial	ships the EC split/label
Niu et al., KeFVP	EMNLP Findings 2023	200, 42.0 code files, pushed 2024-01-28 (2026-09-02)	partial	embeddings as .pkl; not
DialogueGAT	2023	200, 6.0 code files, pushed 2023-06-12 (2026-09-02)	partial	full code but requireme
Yu, Liu & He, Same Company, Same Signal	Findings of ACL 2025	200, 38.0 code files, pushed 2026-08-13 (2026-09-02)	runnable	pickle); torch 1.12 preda
Yang et al., FinTrust	ACL 2023 short	200, 4.0 code files, pushed 2024-01-22 (2026-09-02)	partial	able upstream; not porte
Sound of Risk	2025	200, 0.0 code files, pushed 2025-08-26 (2026-09-02)	none	MIT code, tau<=15 onl
DeFVP	2024	409 (2026-09-02)	none	runs; not ported
ECHO-GL	2024	200, 3.0 code files, pushed 2024-02-26 (2026-09-02)	none	MIT; ships DEC with fea
ECC Analyzer	2024	no_url	none	to 3 decimals
RiskLabs	2024	no_url	none	perturbation harness + 3
NumHTML	AAAI 2022	no_url	none	volatility model
GNA-Vol		no_url	none	repository holds figures
AMA-LSTM		no_url	none	corpus
AT-FinGPT		no_url	none	repository exists but is e
				README states the cod
				volatility
				no code released; GPT-4
				no code released; GPT-4
				paper promises code; no
				artefact URL returns 404
				10-byte stub repository,
				paywalled; no artefact

Table 82: Code-availability audit of the ECC volatility literature (T6R.2; regenerated by `ecvol audit code`). Two of seventeen papers ship code that runs as released; six ship code with a broken environment or missing data; nine ship nothing usable. Probe columns come from the GitHub API on the stated date; verdicts and reasons are curated and re-checked.